

Can Book-to-Market, Size and Momentum Be Risk Factors That Predict Economic Growth?

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Abstract

In this study we examine the extent to which the profitability of the HML, SMB, and WML trading strategies can be linked to macroeconomic risk factors. Using a large cross-section of securities from ten developed markets, we find that the HML and SMB portfolios contain significant information about future GDP growth. The predictive ability of these strategies is to a large degree independent of any information contained in the domestic market factor, which is known to be a leading indicator of economic growth. Even in the presence of popular business cycle variables, HML and SMB retain their ability to predict future economic growth in some of the countries examined. Examination of the portfolio holdings over the life of the trading strategies shows that more than 50% of the securities in the extreme portfolios of HML and SMB remain the same over a period of three years. In most countries, more than 30% of the stocks never exit the extreme portfolios over the whole life of the trading strategy. High book-to-market and small capitalization stocks in these portfolios may be more sensitive to changes in the economic environment because they seem to be entrenched in their respective categories. The vulnerability of high book-to-market and size stocks to changes in the economic environment appears to give rise to the link between the performance of the HML and SMB trading strategies and future economic growth. Little evidence is found to support this hypothesis for the returns of the WML strategy. Finally, industry factors cannot explain the returns of any of the three strategies.

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A growing body of research shows that certain empirical anomalies found in the U.S. stock market appear to also exist in international markets. Fama and French (1998) show that there exists a strong value premium in global stock markets. Rouwenhorst (1998) finds international evidence for a momentum effect. Following Banz (1981), many authors have investigated whether small capitalization stocks outperform large capitalization stocks on a risk-adjusted basis, with varying degrees of success.ⁱ However, researchers have so far found little evidence of a relation between these three return-based anomalies and intuitive economic risk factors. This paper takes one step in that direction by linking the unidentified return-based factors to future growth in the macro economy.

We construct the HML, SMB, and WML portfolio strategies using security returns from ten developed markets. For each market, HML is the return to a portfolio strategy that is long on high book-to-market stocks and short on low book-to-market stocks, holding the other two attributes (size and momentum) constant. Likewise, SMB and WML are returns to long-short portfolios constructed using market capitalization and past year's returns information (momentum), respectively, holding the other two attributes constant.

The core results of the paper reveal that the HML and SMB portfolios are related to future growth in the real economy. This is documented using regression analysis but also by calculating the returns of the trading strategies during both good and bad states of the business cycle in each country. The predictability of HML and SMB is not subsumed when the market factor is included in the regression model. Inclusion of other business

cycle variables does not eliminate the forecasting ability of HML and SMB. We find, however, little evidence of a relation between the returns of the WML strategy and the real economy.

Our results suggest that HML and SMB proxy for underlying risk factors that capture future changes in the economic environment. We track the composition of the portfolios over the life of the strategies and find that the HML and SMB portfolios change substantially less than the WML portfolios over time. Stocks that are classified as high or low book-to-market (B/M) tend to retain this characteristic for several years. A similar observation is made for small and large capitalization stocks. In both cases, about 50% of the securities in the extreme portfolios of HML and SMB remain the same over a period of three years. In most countries, more than 30% of the stocks never exit the extreme portfolios over the life of the trading strategies. This means that the stocks that comprise the HML and SMB portfolios are largely entrenched in their respective categories. This characteristic may make them more vulnerable to changes in future economic conditions than it is the case for the average stock in the market.

Persistently high B/M stocks face a higher risk of distress and they are more likely to survive when the economic outlook is good rather than bad. Similarly, small capitalization stocks are more likely to do well during periods of economic growth, and more likely to be the first to disappear during periods of economic slowdown. It appears that this potential vulnerability of the high B/M and small capitalization stocks gives rise to the link between the performance of the HML and SMB strategies and future economic growth that we document in this study.

Since the B/M and size characteristics of securities are rather persistent over time, the market is aware of them, and investors require an additional risk premium to hold such securities. This premium is captured by the HML and SMB strategies since they are long on high B/M and small capitalization stocks and short on low B/M and large capitalization stocks, respectively. Furthermore, since the performance of HML and SMB is linked to future economic growth, the hypothesis of Fama and French (1992, 1993, 1995, 1998) that they act as state variables in the context of Merton's (1973) Intertemporal Capital Asset Pricing Model (ICAPM) is supported by our results.

Our analysis also confirms previous findings that the HML, SMB and WML zero investment portfolios provide positive returns. In most countries these returns are both economically and statistically significant. The momentum portfolio strategy yields the highest returns of the three portfolio strategies examined. Conspicuously absent is a momentum effect in Japan. Momentum strategies are particularly profitable when the portfolios are rebalanced on a quarterly basis.ⁱⁱ The HML and SMB strategies are more profitable in larger, more liquid markets. Both of these strategies generate lower turnover than the WML strategy, and none of the three yield large positions in any particular industrial sector.

The paper is organized as follows: Section 1 describes the dataset, and details our portfolio construction. Section 2 briefly summarizes the characteristics and performance of the three strategies internationally. In Section 3 we present our results on the relation between the returns on the trading strategies and the macroeconomy. Section 4 examines the composition of the portfolios generated by the trading strategies, by tracking the

turnover of stocks in the portfolios over the life of the strategies. It also examines their industry breakdown. We conclude in Section 5 with a summary of our results.

1.1: Dataset Description

Our data are obtained from Datastream International. They include securities from Australia, Canada, France, Germany, Italy, Japan, the Netherlands, Switzerland, the United Kingdom, and the United States. We use end-of-month prices, dividend yields, price-to-book ratios, and market capitalization data both for currently trading and defunct securities. In the US and the UK, we restrict the number of defunct stocks examined to those that have at least 60 observations of prices.ⁱⁱⁱ We excluded from our database duplicates, cross listings, as well as preferred shares, warrants, and unit trusts.

The number of total securities in our dataset is summarized in Appendix 1. It includes more than 9,700 price series. For each country we present the total number of currently listed stocks, broken down into the currently active stocks (5,640) and the defunct stocks (4,113).

Monthly local currency total returns are calculated by spreading evenly the monthly dividends throughout each year. This method, which represents the only option we had available, may smooth the series to some extent, but it does not affect the means. All returns in our analysis are reported annualized.

As a quality control of our data, we use our database to construct market capitalization weighted country indexes and we compare them with those provided by Morgan Stanley Capital International (MSCI). The results are reported in Appendix 2. The distributional characteristics of the two sets of indexes are very similar. Simple t-

tests fail to reject the hypothesis that the means in the two sets of indexes are equal. Similarly, F-tests cannot reject the hypothesis that their variances are also equal. The advantage of our dataset compared to the constituents of the MSCI indexes used in the Fama and French (1998) study is that our data include also small capitalization stocks, which makes tests of the SMB strategy possible.

1.2. Portfolio Construction

In order to capture all three empirical anomalies in tandem, the HML, SMB, and WML portfolios are constructed as follows.^{iv} For a given country and at each point in time, we use only stocks for which we have the market capitalization (MV), at least twelve monthly observations so as to be able to calculate the momentum, and a positive book-to-market ratio (B/M). We consider only the twelve months momentum strategy and we implement it by calculating the average of past year's returns, excluding the most recent month (ave12).^v

To construct the portfolios, we sort all stocks that pass the above requirements by B/M and create tritile portfolios. We then take the portfolio of stocks with the highest B/M's and re-sort all stocks by MV, thereby creating three MV portfolios within the high B/M group. We repeat the same procedure for the medium B/M and low B/M groups. After sorting for B/M and MV, we have nine portfolios. We then sort the securities in each of these nine portfolios according to ave12 and create tritile portfolios within the nine portfolios. We obtain in this manner twenty-seven portfolios.

Figure 1 depicts the portfolio construction procedure. "Losers" are the bottom third of the total stocks with the lowest last year's average returns, excluding the most

recent month. “Winners” are the top third of the total stocks with the highest last year’s average returns, excluding the most recent month and “Medium” contain the remaining third of the stocks.

Figure 1: Portfolio Construction Procedure

Book-to-Market	Market Capitalization	Past Year’s Returns	Portfolio
High B/M	Small MV	Losers AVE12	P1
		Medium AVE12	P2
		Winners AVE12	P3
	Medium MV	Losers AVE12	P4
		Medium AVE12	P5
		Winners AVE12	P6
	Big MV	Losers AVE12	P7
		Medium AVE12	P8
		Winners AVE12	P9
Medium B/M	Small MV	Losers AVE12	P10
		Medium AVE12	P11
		Winners AVE12	P12
	Medium MV	Losers AVE12	P13
		Medium AVE12	P14
		Winners AVE12	P15
	Big MV	Losers AVE12	P16
		Medium AVE12	P17
		Winners AVE12	P18
Low B/M	Small MV	Losers AVE12	P19
		Medium AVE12	P20
		Winners AVE12	P21
	Medium MV	Losers AVE12	P22
		Medium AVE12	P23
		Winners AVE12	P24
	Big MV	Losers AVE12	P25
		Medium AVE12	P26
		Winners AVE12	P27

The three trading strategies are performed as follows:

$$HML = 1/9 * ((P1-P19)+(P2-P20)+(P3-P21)+(P4-P22)+(P5-P23)+(P6-P24)+(P7-P25)+(P8-P26)+(P9-P27))$$

$$SMB = 1/9 * ((P1-P7)+(P2-P8)+(P3-P9)+(P10-P16)+(P11-P17)+(P12-P18)+(P19-P25)+(P20-P26)+(P21-P27))$$

$$WML = 1/9 * ((P3-P1)+(P6-P4)+(P9-P7)+(P12-P10)+(P15-P13)+(P18-P16)+(P21-P19)+(P24-P22)+(P27-P25))$$

where portfolios P1 to P27 are defined in Figure 1.

HML represents the return to a portfolio that is long on high B/M stocks and short on low B/M stocks, controlling for the size and momentum effects. In other words, HML is a zero investment strategy that is both size and momentum neutral. Similar interpretations can be given for SMB and WML.

The twenty-seven portfolios are value-weighted at construction. Given that our database includes also small capitalization securities, constructing value-weighted rather than equally weighted portfolios results in more realistic returns for the trading strategies. The returns for the portfolio strategies are calculated for quarterly, semi-annual, and annual re-balancing frequencies.^{vi}

Our annual sorts are similar to those of Fama and French (1992). We use December-end B/M values, June-end market capitalization, and past twelve months of returns subsequent to July. In addition, for the United States we exclude all financial securities. We differ from Fama and French (1992) in that we perform one additional sort on ave12. For the semi-annual re-balancing portfolios, we use B/M available six months prior to the portfolio re-balancing date. We re-balance semi-annual portfolios at June-end and December-end. For the quarterly re-balancing portfolios, we use the prior three-month's B/M values. The quarterly sorted portfolios are re-balanced at the end of March, June, September, and December. Shorter horizon re-balancing strategies are akin to more aggressive use of B/M information. If a stock does not have returns for any month through the duration of the holding period, we invest that portion of the portfolio into the corresponding country's risk-free asset.

In Canada, the United Kingdom, and the United States, we have portfolio returns over the whole sample period, July 1976 to May 1997. Depending on the country, however, the series of portfolio returns may be shorter.

To provide an idea of the number of stocks used in the portfolios over the life of the strategies, Appendix 3 summarizes the number of securities used in annual portfolio sorts. It discloses how many stocks are used in the portfolios each year, and the length of time during which the trading strategies were performed. Note that for Italy, the Netherlands, and Switzerland, the number of securities is much smaller than for the other countries. This means that since our portfolio construction procedure results in the creation of 27 portfolios, some of the portfolios for these three countries may contain as few as 1 to 3 stocks. Therefore, the results for these countries should be interpreted with caution, especially those referring to the relationship of HML, SMB, and WML with macroeconomic growth.

2. Characteristics and Performance of the HML, SMB, and WML Strategies Internationally

Before we test for the potential link between the returns of the trading strategies and macroeconomic growth, it is useful to examine the performance of these strategies in the markets and time period covered by our data. Table 1 reports the returns of the three strategies for annual, semi-annual, and quarterly holding periods.

Our results confirm those of Fama and French (1998) that the value premium is pervasive. We find that, in general, the returns of HML are higher when the portfolios are

rebalanced more frequently. For quarterly re-balancing, nine out of ten countries have statistically positive returns.

We also find that the momentum strategy examined provides significantly positive returns. This is consistent with Rouwenhorst's (1998) findings.^{vii} It appears that momentum is very sensitive to the re-balancing period. Returns to WML decrease rapidly as the re-balancing interval increases. Note that the momentum strategy is not profitable in Italy and Japan.

Finally, we verify the existence of a size premium in our larger countries. SMB produces statistically significant returns in Canada, France, Japan, the United States and marginally significant returns in the United Kingdom. It does not appear to be profitable, however, in countries whose markets are typically smaller, less liquid and dominated by a few large capitalization stocks.

3. The Relation between the Returns on the Trading Strategies and the Macroeconomy

This section examines the potential relation between the return on HML, SMB, and WML strategies and future economic growth. Our analysis makes use of additional data, which are discussed below.

3.1. Country Variables

Apart from individual security returns, we use in each country, the return on the market portfolio (MKT), the dividend yield (DY), the short-term interest rate (TB), the term

spread (TERM), the growth in the Gross Domestic Product (GDP) and the Industrial Production (IDP). Appendix 5 provides summary statistics.

For a given country, DY represents the dividend yield index created from our database. The dividend index is the ratio of dividends from a market-capitalization-weighted portfolio for the past year to the value of the portfolio at quarter-end. The short-term interest rates and the long-term bond yields are from the International Monetary Fund, and they are reported on an annual basis. For Australia, Canada, Switzerland, the United States, and the United Kingdom we use the three-month Treasury Bill (TB). For the remaining countries we use the Call Money Rate. Both the Treasury bill yield and the money rate are the average of daily rates from the last month of the quarter. We create the variable TERM as the difference between the long-term government bond yield and the TB for each country. TERM measures the slope of the country yield curve. The GDP and IDP series are obtained from the OECD Main Indicators and the National Government Series. We compute continuously compounded quarterly growth rates of each country's GDP and IDP, and report the annualized mean and standard deviation.^{viii}

3.2. Returns on the Trading Strategies at Different States of the Macroeconomy

We examine the returns on the HML, SMB, and WML strategies at different states of future economic growth. In particular, we associate next year's growth in GDP with past year's annual return, and we subsequently sort by growth in GDP every quarter. We call "Good States" of the economy those states that exhibit the highest 25% of future GDP growth, and "Bad States" those with the lowest 25% of future GDP growth.

The results are presented in Table 2. They show that the returns on HML and SMB are positively related to future growth in the macroeconomy. High portfolio returns precede periods of high GDP growth. Similarly, low portfolio returns are associated with low future GDP growth. This is true for the HML strategy in eight out of the ten countries, whereas it holds for nine out of ten countries for SMB, and six out of ten countries for WML.^{ix}

3.3. The Relation between the Return on the Trading Strategies and the Macroeconomy using Regression Analysis

3.3.1. Univariate Regressions

The above results, regarding the relation of the three strategies and economic growth, are confirmed in this section using regression analysis. Table 3 reports results from univariate regressions of future growth in GDP on past holding period's returns in HML, SMB, and WML. The first panel reports results from regressions that use quarterly holding period returns while the middle and last panels report results from regressions that use semi-annual and annual holding period returns respectively. All regressions use quarterly data and they are of the form:

$$\text{GDPgrowth}_{(t,t+4)} = a + b * \text{FactorRet}_{(t-n,t)} + e_{(t,t+4)}$$

- n : 1 quarter, 2 quarters, and 4 quarters;
- GDPgrowth : Seasonally adjusted continuously compounded growth rates for country GDP;
- FactorRet : MKT, HML, SMB, and WML month returns compounded to 1 quarter, 2 quarters or 4 quarters; and
- $e_{(t,t+4)}$: The residual term of the regression. It has zero expectation and is uncorrelated with the factor returns.

GDP growth rates are observed at quarterly frequencies, and therefore, consecutive semi-annual and annual growth rates have one and three overlapping quarters, respectively. This induces serial correlation in the residuals of our regressions. To correct for this, we use the Newey and West (1987) estimator and set the parameter q equal to one for the semi-annual holding period regressions and three for the annual holding period regressions.^x

In most countries, the returns of the three strategies have greater predictive power over future GDP growth as the holding period increases from one quarter to four quarters. We would expect the slope coefficients to be positive if high returns of these strategies are associated with future good states of the economy. That would mean that high B/M and small capitalization stocks are better able to prosper when periods of high economic growth are expected, and they face a higher risk of financial distress if economic growth slows down. Indeed, in the annual holding period regressions, the coefficient of HML is positive for eight of the ten countries. For SMB and WML, the coefficients are positive in nine and six countries respectively. However, only one of these coefficients is statistically significant in the case of the WML strategy, compared to three for HML and seven for SMB. The returns of the WML strategy appear to contain little, if any, information about future economic growth.

It is known that the market factor is a leading indicator of future economic growth. Our results confirm this stylized fact. For the annual holding period regressions, the slope coefficient is positive in all ten countries which means that periods of high market returns are followed by periods of high economic growth. Furthermore, the coefficients are of similar order of magnitude as those of the HML and SMB portfolios.

One would be tempted to infer that HML and SMB capture the same information as MKT. This is not the case, however. Appendix 4 reports regression betas of HML, SMB, and WML with the market factor. They vary between -0.331 and 0.240 . Therefore, the ability of HML and SMB to explain future economic growth is at least to some extent independent of that of the market factor. This is confirmed by the results of the following section.

3.3.2. Multiple Regressions that include the Market Factor

In this section we explicitly compare the informational content of HML and SMB with that of MKT. Since GDP growth changes little from one quarter to another, we concentrate on annual holding period regressions, which are more informative.

The bivariate regressions we estimate include the market factor and the return on a trading strategy:

$$\text{GDPgrowth}_{(t,t+4)} = a + b \cdot \text{MKT}_{(t-4,t)} + c \cdot \text{FactorRet}_{(t-4,t)} + k_{(t,t+4)}$$

MKT : Annualized monthly excess market return over the risk-free rate.
FactorRet : Annualized monthly returns on HML, SMB, and WML.
 $k_{(t,t+4)}$: the residuals of the regression.

Table 4 reports the results. Even in the presence of the market factor, the coefficients of HML and SMB remain positive and statistically significant. Furthermore, their magnitude is largely unchanged. HML has a positive and statistically significant coefficient in Australia, France, Italy, United Kingdom, and USA. The coefficient of SMB is positive and significant in Australia, Canada, France, Germany, Italy, the Netherlands, and the United Kingdom. As expected, the adjusted R-squares are now larger than those from the univariate regressions of Section 3.3.1 that included only the

market factor. Therefore, the returns of HML and SMB contain information about the future state of the macroeconomy, over and above the information contained in the market factor. Once again, WML appears to have virtually no ability to explain future economic growth.

We now run a “horse-race” between HML and SMB by estimating the following regression model:

$$\text{GDPgrowth}_{(t,t+4)} = a + b \cdot \text{MKT}_{(t-4,t)} + c \cdot \text{HML}_{(t-4,t)} + d \cdot \text{SMB}_{(t-4,t)} + u_{(t,t+4)}$$

Where $u_{(t,t+4)}$ denotes the residuals of the regression.

Table 5 reports the results. For seven out of the ten countries, either HML or SMB remains statistically significant. A comparison of the coefficients in Table 4 and Table 5 reveals that the sign and the magnitude of the coefficients remain relatively stable in the presence of both factors. In France HML subsumes the effects of SMB. The opposite occurs in Australia and United Kingdom where SMB subsumes the effects of HML. It seems that the informational content of HML and SMB regarding the future state of the macroeconomy is to some extent country specific. This is a plausible result, given that the countries included in our analysis differ in terms of the size of their markets, the average market capitalization, and their accounting standards.

3.3.3. Regressions That Include Business Cycle Variables and the Market Factor

Given the results of the previous sections, it is instructive to examine how much of the information contained in the returns of the trading strategies regarding future economic growth is also present in popular business cycle variables. The multiple regressions estimated are of the form:

$$\text{GDPgrowth}_{(t,t+4)} = a + b \cdot \text{MKT}_{(t-4,t)} + c \cdot \text{FactorRet}_{(t-4,t)} + d \cdot \text{TB}_{(t)} + f \cdot \text{DY}_{(t)} + g \cdot \text{TERM}_{(t)} + h \cdot \text{IDPgrowth}_{(t-4,t)} + q_{(t,t+4)}$$

$\text{TB}_{(t)}$: Annual Treasury bill yield or Call Money Rate;
 $\text{DY}_{(t)}$: Dividend yield on country market capitalization index;
 $\text{TERM}_{(t)}$: 10 year government yield minus $\text{TB}_{(t)}$; and
 $\text{IDPgrowth}_{(t-4,t)}$: Past one-year growth in country industrial production.
 $q_{(t,t+4)}$: the residuals of the regression.

Table 6A presents the results. It appears that even in the presence of business cycle variables in addition to the market factor, HML still maintains a positive and statistically significant relation with future economic growth in France, Italy, and the United Kingdom (see first panel). Furthermore, the predictive ability of SMB remains significant in France, Germany, Italy, the Netherlands, and the United States whereas it is subsumed by the presence of business cycle variables in Australia, Canada, Germany, and the United Kingdom (see second panel). Finally, WML has a positive and significant coefficient in the US and the UK.

We also perform regressions that include both the HML and SMB variables in addition to the business cycle state variables. The results are reported in Table 6B. HML has now a positive and significant coefficient only in France and the UK, whereas SMB has a positive and significant coefficient in France, Germany, Netherlands and the US. There is obviously some overlap in the information content of the returns of HML, SMB, and the business cycle variables. Our aim in this section has not been to propose a new model for predicting future economic growth. Instead, we aimed to evaluate the level and nature of information contained in the returns of the trading strategies regarding future economic growth. Our results show that at least HML and SMB contain significant

information regarding future growth in the macroeconomy. Furthermore, some of this information is similar in nature to that contained in popular business cycle variables. Our findings support the Fama and French (1992, 1993, 1995, 1996, 1998) hypothesis. They imply that it is reasonable to consider HML and SMB as state variables that predict future changes in the investment opportunity set, in the context of Merton's intertemporal capital asset pricing model.

4. The Composition of the HML, SMB, and WML Portfolios

The ability of the returns of the trading strategies to predict future economic growth warrants a closer look at the composition of the portfolios that give rise to these trading strategies. In particular, it is interesting to examine whether the B/M and size characteristics of the stocks in those portfolios are persistent over time. Stocks with persistently high B/M, and therefore a high risk of financial distress, are more likely to do well when the outlook for the economy is good rather than bad. Similarly, small capitalization stocks that remain small for a prolonged period of time, because of lack of growth opportunities, are more likely to do well during periods of high economic growth.

To verify whether the above hypothesis is reasonable, we track the turnover of the portfolios over the whole life of the trading strategies. Specifically, we calculate the percentage of stocks that remain the same in the extreme portfolios (i.e., high B/M, low B/M, small capitalization, large capitalization, winners, and losers) over successive years. We also report the percentage of stocks that disappear from the portfolios, either because the stocks become defunct or because they are merged with other stocks.

Table 7 reports the results. In general, more than 50% of the securities in the extreme portfolios of the HML and SMB strategies remain the same over a period of three years, and more than 30% of the stocks never exit the extreme portfolios over the whole life of the trading strategies. Note, however, that the turnover for the US is somehow higher. In particular, 34% of the high B/M stocks and 46% of the low B/M stocks remain the same over a three-year period. In the case of the SMB strategy, 42% of the small capitalization stocks and 71% of the large capitalization stocks do not exit the portfolio over the three-year period. In contrast to the results for the HML and SMB strategies, the turnover for the WML strategy is much higher for all countries. Only about 30% of the stocks remain the same in a period of three years.

The above results suggest that the book-to-market and size characteristics of the securities included in the extreme portfolios are rather stable over time. This stability may induce the link between the performance of these strategies and future macroeconomic growth. Consistent with our hypothesis are our findings for the WML strategy. The lack of a link between the returns of the WML strategy and economic growth is accompanied by a relatively high turnover of stocks in the extreme portfolios.

Skeptics of our hypothesis may argue that the returns of the trading strategies can be due to industry factors captured by the construction of the portfolios. It is possible that stocks that belong to different industries differ in their B/M ratios. In that case, by going long on high B/M stocks and short on low B/M stocks, the HML strategy may load on a particular industry factor. A similar interpretation could be provided for the SMB and WML strategies.

We use the Datastream Level 3 industrial codes to classify stocks into the following industries: Consumer Goods, Financials, General Industrials, Resources, Services, Utilities, and Others. We then compare the industry breakdown of our sample with that of the annually rebalanced extreme portfolios.^{xi}

The results are presented in Table 8. No concentration to a particular industry emerges from the comparison. In all three strategies, the industrial base of each market seems to be fairly represented in terms of the number of securities participating in the trading strategies. Therefore, it is unlikely that the performance of HML, SMB, and WML is due to industry factors.

5. Conclusions

The aim of this study was to investigate the extent to which the profitability of the HML, SMB, and WML trading strategies can be linked to macroeconomic risk factors.

Using a large cross-section of securities from ten developed markets, we found that at least the HML and SMB portfolios contain significant information about future GDP growth. The apparent predictive ability of these strategies appears to be to a large degree independent of any information contained in the market factor, which is known to be a leading indicator of economic growth. Even in the presence of popular business cycle variables, HML and SMB retain their ability to predict future economic growth in some of the countries examined.

The portfolio holdings of the HML and SMB trading strategies remain relatively stable over time. In general, more than 50% of the securities in the extreme portfolios of HML and SMB remain the same over a period of three years. In most countries, more

than 30% of the stocks never exit the extreme portfolios over the whole life of the trading strategy. It appears, therefore, that high B/M and small capitalization stocks are entrenched in their respective categories, which in turn induces a positive relation between the returns of these strategies and future economic growth. Stocks with persistently high B/M ratios face a higher risk of financial distress, and they are more likely to do well in periods of high economic growth. By the same token, small capitalization stocks that remain in this category for a prolonged period of time, as a result of lack of growth opportunities, are more likely to prosper during periods of good economic outlook.

The results of this study suggest that a risk-based explanation for the returns of HML and SMB strategies is plausible and likely. Fama and French (1993, 1995, 1996, 1998) argue that HML and SMB are state variables that predict future changes in the investment opportunity in the context of Merton's (1973) intertemporal asset pricing model. Our findings support this hypothesis. Finally, our analysis reveals little evidence to support a similar explanation for the returns of WML.

Bibliography

- Asness, Clifford S., 1995, "The Power of Past Stock Returns to Explain Future Stock Returns," Working Paper, Applied Quantitative Research.
- Banz, Rolf W., 1981, "The Relationship Between Return and Market Value of Common Stocks," *Journal of Financial Economics* 9, 3-18.
- Carhart, Mark M., 1997, "On Persistence in Mutual Fund Performance," *Journal of Finance* 52, 57-82.
- Carhart, Mark M., Robert J. Krail, Ross L. Stevens, and Kelly D. Welch, 1995, "Testing the conditional CAPM," Manuscript, University of Chicago.
- Chen, Nai-Fu, 1991, "Financial Investments Opportunities and the Macroeconomy," *Journal of Finance* 46, 529-554.
- Chan, Louis K., Yasushi Hamao, and Josef Lakonishok, 1991, "Fundamentals and Stock Returns in Japan," *Journal of Finance* 46, 1739-1789.
- Fama, Eugene F., 1981, "Stock returns, real activity, inflation, and money," *American Economic Review* 71, 545-565.
- Fama, Eugene F., 1990, "Stock Returns, Expected Returns, and Real Activity," *Journal of Finance* 35, 1089-1108.
- Fama, Eugene F. and Kenneth R. French, 1992, "The Cross-Section of Expected Stock Returns," *Journal of Finance* 47, 427-465.
- Fama, Eugene F. and Kenneth R. French, 1993, "Common Risk Factors in the Returns on Stocks and Bonds," *Journal of Financial Economics* 33, 3-56.
- Fama, Eugene F. and Kenneth R. French, 1995, "Size and Book-to-Market Factors in Earnings and Returns," *Journal of Finance* 50, 131-155.
- Fama, Eugene F. and Kenneth R. French, 1996, "Multifactor Explanations of Asset Pricing Anomalies," *Journal of Finance* 51, 55-184.
- Fama, Eugene F. and Kenneth R. French, 1998, "Value versus Growth: The International Evidence", *Journal of Finance*, 1975-1998.
- Hawawini, Gabriel and Donald B. Kiem, 1995, "On the Predictability of Common Stock Returns: World-Wide Evidence," *Handbooks in OR & MS, Vol.9: Finance*, Elsevier Science B.V.

Heston, Steven L., K. Geert Rouwenhorst, and Roberto E. Wessels, 1995, "The Structure of International Stock Returns and the Integration of Capital Markets", *Journal of Empirical Finance*, 2, 173-197.

Jegadeesh, Narasimhan, and Sheridan Titman, 1993, "Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency," *Journal of Finance* 48, 65-91.

Loughran, Tim, 1997, "Book-to-Market Across Firm Size, Exchange, and Seasonality: Is There an Effect?", *Journal of Financial and Quantitative Analysis*, 32(3), 249-268.

Merton, Robert C., 1973, "An Intertemporal Capital Asset Pricing Model," *Econometrica* 41, 867-887.

Newey, W. and K. West, 1987, "A Simple Positive-Definite Heteroskedasticity and Autocorrelation Consistent Covariance Matrix," *Econometrica* 55, 703-708.

Rouwenhorst, K. Geert, 1998, "International Momentum Strategies", *Journal of Finance*.

Sharpe, William F., 1964, "Capital Asset Pricing: A Theory of Market Equilibrium Under Conditions of Risk," *Journal of Finance* 19, 425-442.

White, Halbert, 1980, "A Heteroskedasticity-Consistent Covariance Matrix Estimator and Direct Test for Heteroskedasticity," *Econometrica* 48, 817-838.

Table 1
Summary of Long-Short Portfolios Annual Returns
Portfolio Construction Over Different Horizons

Country	Quarterly Rebalancing			Semi-Annual Rebalancing			Annual Rebalancing		
	mean	std.	t-stat.	mean	std.	t-stat.	mean	std.	t-stat.
Australia									
HML	10.10%	13.59%	2.64**	9.14%	14.06%	2.29**	7.02%	13.14%	1.85*
SMB	7.50%	16.20%	1.65	2.79%	16.06%	0.61	8.82%	19.90%	1.53
WML	8.32%	15.24%	1.94*	12.56%	15.35%	2.88**	4.18%	15.74%	0.92
Canada									
HML	10.22%	11.62%	3.90**	8.56%	10.69%	3.53**	4.00%	10.71%	1.63
SMB	4.57%	10.37%	1.96*	6.02%	10.79%	2.46**	6.64%	10.43%	2.77**
WML	16.02%	13.81%	5.14**	10.52%	14.22%	3.26**	1.69%	13.22%	0.56
France									
HML	13.05%	9.71%	5.06**	12.05%	9.26%	4.85**	7.00%	8.74%	2.88**
SMB	5.59%	10.91%	1.93*	5.46%	11.42%	1.79*	5.57%	10.33%	1.94*
WML	12.58%	9.43%	5.02**	8.57%	9.14%	3.50**	7.64%	9.15%	3.00**
Germany									
HML	6.01%	6.75%	2.87**	3.14%	5.96%	1.66*	2.74%	7.59%	1.14
SMB	0.67%	9.70%	0.22	0.82%	9.63%	0.27	1.39%	9.55%	0.46
WML	9.93%	9.31%	3.44**	10.08%	9.62%	3.30**	7.70%	9.83%	2.47**
Italy									
HML	8.41%	10.20%	2.63**	7.47%	9.24%	2.55**	5.18%	10.27%	1.59
SMB	1.44%	10.36%	0.44	2.92%	10.32%	0.89	0.68%	10.93%	0.20
WML	5.64%	12.83%	1.40	3.24%	11.89%	0.86	-2.77%	12.18%	-0.72
Japan									
HML	12.03%	9.40%	5.26**	6.85%	9.77%	2.84**	8.21%	9.06%	3.61**
SMB	6.10%	15.13%	1.66*	6.92%	15.37%	1.82*	6.93%	14.08%	1.96*
WML	0.75%	10.81%	0.29	-1.42%	11.30%	-0.51	-3.09%	10.38%	-1.19
Netherlands									
HML	-0.11%	11.59%	-0.04	0.96%	11.60%	0.36	-1.22%	12.08%	-0.43
SMB	-0.15%	12.52%	-0.05	1.82%	12.73%	0.62	-1.75%	11.59%	-0.64
WML	13.94%	12.85%	4.75**	9.61%	12.85%	3.25**	11.03%	11.92%	3.92**
Switzerland									
HML	6.41%	10.53%	2.04**	7.62%	10.57%	2.38**	5.85%	9.90%	1.86*
SMB	-2.42%	10.14%	-0.80	-3.39%	11.01%	-1.02	-3.60%	9.19%	-1.23
WML	8.76%	9.99%	2.93**	7.54%	10.04%	2.48**	8.04%	8.97%	2.82**
United Kingdom									
HML	8.58%	6.39%	5.96**	6.92%	5.84%	5.22**	4.85%	5.86%	3.60**
SMB	4.28%	11.40%	1.67*	4.09%	11.02%	1.63	4.38%	10.94%	1.74*
WML	15.40%	6.67%	10.24**	13.23%	6.70%	8.69**	9.08%	6.24%	6.33**
United States									
HML	8.31%	9.23%	4.00**	6.82%	9.00%	3.34**	5.08%	8.74%	2.53**
SMB	9.32%	11.05%	3.74**	8.64%	11.44%	3.33**	6.84%	10.74%	2.77**
WML	11.45%	9.57%	5.31**	9.78%	9.34%	4.61**	4.39%	8.99%	2.12**

* Significant at the 10% Level.

** Significant at the 5% Level.

HML is the return on a portfolio that is long on high book-to-market stocks and short on low book-to-market stocks, holding the size and momentum characteristics of the portfolio constant. SMB is the return on a portfolio that is long on small capitalization stocks and short on large capitalization stocks, holding the book-to-market and momentum characteristics of the portfolio constant. WML is the return on a portfolio that is long on the best performing stocks of the past year ("winners") and short on the worst performing stocks of the past year ("losers"), holding the book-to-market and size effects of the portfolio constant. Returns and standard deviations are reported annualized.

Table 2: The Performance of the HML, SMB, and WML Strategies
at Good, Bad, and Normal States of the Economy

Past Year Return on Return-Based Factor Sorted by Future GDP Growth									
Country	HML			SMB			WML		
	Good States	Normal	Bad States	Good States	Normal	Bad States	Good States	Normal	Bad States
Australia	23.88%	6.39%	8.77%	22.93%	6.32%	-1.87%	5.75%	10.80%	5.15%
Canada	8.19%	9.57%	15.57%	9.15%	6.25%	-0.41%	21.53%	20.02%	11.56%
France	30.11%	14.40%	3.01%	16.29%	5.15%	-0.94%	6.74%	19.77%	9.43%
Germany	9.39%	6.02%	2.80%	13.47%	-1.42%	-2.44%	17.50%	10.80%	8.53%
Italy	19.75%	10.92%	-0.92%	5.71%	3.73%	-6.98%	6.10%	4.56%	6.35%
Japan	13.02%	15.60%	10.07%	12.98%	8.25%	2.35%	1.79%	3.52%	-5.06%
Netherlands	10.68%	-1.41%	-2.46%	9.61%	0.23%	-8.09%	6.69%	15.20%	18.90%
Switzerland	5.91%	6.58%	7.85%	1.83%	-9.69%	3.91%	13.52%	8.02%	7.51%
United Kingdom	17.85%	7.84%	3.38%	13.59%	6.58%	-7.10%	17.22%	15.07%	17.23%
United States	8.49%	12.58%	2.81%	12.98%	7.20%	10.03%	13.82%	11.43%	10.71%

HML is the return on a portfolio that is long on high book-to-market stocks and short on low book-to-market stocks, holding the size and momentum characteristics of the portfolio constant. SMB is the return on a portfolio that is long on small capitalization stocks and short on large capitalization stocks, holding the book-to-market and momentum characteristics of the portfolio constant. WML is the return on a portfolio that is long on the best performing stocks of the past year (“winners”) and short on the worst performing stocks of the past year (“losers”), holding the book-to-market and size effects of the portfolio constant. GDP growth is calculated as the continuously compounded growth rate in a country’s Gross Domestic Product, which is seasonally adjusted. For Japan we use the seasonally adjusted Gross National Product. We characterize as “Good States” of the economy, those states that exhibit the highest 25% of future GDP growth, and “Bad States” those with the lowest 25% of future GDP growth. The middle 50% is characterized as “Normal” states of the economy.

Table 3: Univariate Regressions of GDP growth rates Conditional on Past One-Quarter, Two-Quarters, and Four-Quarters of Factor Returns
 $GDPgrowth(t,t+4) = a + b * FactorRet(t-n,t) + e(t,t+n)$ where $n=1,2$, or 4.

Country	Time Period	Slope Coefficient				t-statistics				Coefficient of Determinations			
		MKT	HML	SMB	WML	MKT	HML	SMB	WML	MKT	HML	SMB	WML
		Past One Quarter GDPgrowth(t,t+4) = a + b *FactorRet (t-1,t) + e(t,t+4)											
Australia	1984Q4-1996Q2	0.047	0.019	0.051	0.021	1.28	0.48	2.04**	0.66	3.9%	-1.7%	4.8%	-1.7%
Canada	1977Q4-1996Q2	0.133	-0.017	0.048	0.009	3.03**	-0.34	1.55	0.18	15.7%	-1.2%	0.0%	-1.3%
France	1983Q2-1996Q2	0.016	0.098	0.110	-0.010	0.73	4.60**	4.05**	-0.28	-0.7%	11.1%	12.8%	-1.8%
Germany	1987Q1-1996Q3	-0.011	-0.239	0.189	0.095	-0.17	-1.24	3.08**	0.78	-2.6%	2.9%	6.3%	-0.7%
Italy	1987Q2-1996Q3	0.031	0.103	0.124	-0.022	1.19	2.72**	3.10**	-0.80	2.0%	10.4%	13.5%	-1.5%
Japan	1980Q3-1996Q3	0.018	0.018	0.048	0.030	0.72	0.52	1.52	1.03	-0.6%	-1.3%	2.3%	-0.7%
Netherlands	1978Q2-1996Q3	0.059	-0.006	0.015	-0.048	2.17**	-0.14	0.58	-1.44	5.3%	-1.3%	-1.1%	1.4%
Switzerland	1986Q2-1995Q4	0.010	-0.029	-0.053	-0.010	0.30	-1.00	-1.05	-0.49	-2.3%	-1.3%	2.1%	-2.5%
UK	1977Q4-1996Q3	0.070	0.161	0.137	-0.043	1.95*	2.39**	3.27**	-0.69	4.5%	7.0%	12.5%	-0.9%
USA	1977Q4-1996Q3	0.084	0.001	0.069	0.105	2.01**	0.02	1.66*	1.52	7.2%	-1.4%	2.8%	3.0%
Past Two Quarters GDPgrowth(t,t+4) = a + b *FactorRet (t-2,t) + e(t,t+4)													
Australia	1985Q1-1996Q2	0.048	0.022	0.054	0.022	1.47	0.75	2.16**	0.86	9.8%	-0.7%	11.5%	-1.0%
Canada	1978Q1-1996Q2	0.098	-0.027	0.048	0.017	3.28**	-0.66	2.00**	0.47	23.0%	-0.4%	2.0%	-0.7%
France	1983Q3-1996Q2	0.014	0.096	0.090	-0.008	0.82	5.84**	5.33**	-0.29	0.3%	28.3%	20.6%	-1.7%
Germany	1987Q2-1996Q3	0.002	-0.089	0.174	0.114	0.03	-0.55	3.66**	1.03	-2.8%	-1.5%	14.7%	3.3%
Italy	1987Q3-1996Q3	0.031	0.087	0.096	-0.013	1.44	2.97**	3.61**	-0.59	7.8%	19.4%	21.1%	-2.1%
Japan	1980Q4-1996Q3	0.024	0.000	0.040	0.041	1.16	0.02	1.72*	1.74*	2.1%	-1.6%	4.1%	1.3%
Netherlands	1978Q3-1996Q3	0.046	0.006	0.055	-0.057	2.41**	0.25	2.15**	-2.41**	8.1%	-1.3%	5.0%	6.6%
Switzerland	1986Q3-1995Q4	0.014	-0.040	-0.053	-0.019	0.47	-1.56	0.60	-0.79	-1.2%	1.0%	8.4%	-2.0%
UK	1978Q1-1996Q3	0.064	0.111	0.109	-0.036	2.29**	2.61**	4.22**	-0.83	8.6%	11.1%	20.7%	-0.6%
USA	1978Q1-1996Q3	0.069	0.010	0.062	0.073	2.23**	0.27	1.52	1.39	12.7%	-1.2%	4.8%	3.5%
Past Four Quarters GDPgrowth(t,t+4) = a + b *FactorRet (t-4,t) + e(t,t+4)													
Australia	1985Q3-1996Q2	0.028	0.036	0.053	0.023	1.31	1.60	3.16**	0.97	7.5%	7.3%	31.9%	0.5%
Canada	1978Q3-1996Q2	0.051	-0.033	0.052	0.021	2.93**	-1.21	2.67**	1.05	16.3%	1.9%	8.2%	1.4%
France	1984Q1-1996Q2	0.009	0.072	0.052	0.001	0.77	4.80**	3.09**	0.03	0.1%	41.4%	16.0%	-2.1%
Germany	1987Q4-1996Q3	0.056	0.070	0.125	0.144	1.40	0.73	3.40**	1.31	6.4%	-1.0%	11.9%	14.9%
Italy	1988Q1-1996Q3	0.031	0.067	0.061	-0.007	2.75**	3.30**	3.06**	-0.54	25.4%	33.0%	20.4%	-2.7%
Japan	1981Q2-1996Q3	0.031	0.006	0.022	0.053	2.04**	0.32	1.40	2.65**	12.3%	-1.5%	2.2%	8.8%
Netherlands	1979Q1-1996Q3	0.029	0.016	0.075	-0.039	1.90*	0.89	4.65**	-1.89*	7.6%	0.7%	20.0%	7.9%
Switzerland	1987Q1-1995Q4	0.016	-0.026	-0.056	-0.001	0.86	-1.68*	-1.88*	-0.06	1.4%	0.6%	27.3%	-2.9%
UK	1978Q3-1996Q3	0.069	0.068	0.070	-0.001	3.29**	2.37**	4.04**	-0.04	19.3%	12.2%	21.2%	-1.4%
USA	1978Q3-1996Q3	0.047	0.024	0.036	0.065	2.33**	1.05	1.06	1.42	11.8%	0.6%	3.4%	5.7%

* Significant at the 10% Level.

** Significant at the 5% Level.

T-statistics are corrected for heteroskedasticity and serial correlation (of lags zero (first panel), one (second panel) and three (third panel)) using the Newey West (1987) estimator. In the regression notation, "FactorRet" stands for MKT, HML, SMB, and WML. MKT is the excess return on the local market index. HML is the return on a portfolio that is long on high book-to-market stocks and short on low book-to-market stocks, holding the size and momentum characteristics of the portfolio constant. SMB is the return on a portfolio that is long on small capitalization stocks and short on large capitalization stocks, holding the book-to-market and momentum characteristics of the portfolio constant. WML is the return on a portfolio that is long on the best performing stocks of the past year ("winners") and short on the worst performing stocks of the past year ("losers"), holding the book-to-market and size effects of the portfolio constant. GDP growth is calculated as the continuously compounded growth rate in a country's Gross Domestic Product, which is seasonally adjusted. For Japan we use the seasonally adjusted Gross National Product. All returns are annualized and continuously compounded.

Table 4: Predicting Annual GDP Growth Rates Conditional on Information about the Return on the Market and HML, SMB, and WML Trading Strategies :
Bivariate Regressions

$$\text{GDPgrowth}(t,t+4) = a + b \cdot \text{MKT}(t-4,t) + c \cdot \text{FactorRet}(t-4,t) + e(t,t+4)$$

Country	Time Period	MKT		HML		adj. R2
		slope	t-stat.	slope	t-stat.	
Australia	1985Q3-1996Q2	0.029	1.39	0.037	2.22**	15.84%
Canada	1978Q3-1996Q2	0.049	2.83**	-0.022	-1.03	16.66%
France	1984Q1-1996Q2	-0.008	-1.16	0.077	5.44**	41.47%
Germany	1987Q4-1996Q3	0.054	1.39	0.025	0.25	3.80%
Italy	1988Q1-1996Q3	0.019	1.86*	0.050	2.44**	38.93%
Japan	1981Q2-1996Q3	0.031	2.10**	0.010	0.62	11.31%
Netherlands	1979Q1-1996Q3	0.035	2.69**	0.026	1.43	11.41%
Switzerland	1987Q1-1995Q4	0.025	1.32	-0.042	-2.10**	6.52%
United Kingdom	1978Q3-1996Q3	0.067	4.05**	0.065	2.74**	30.50%
United States	1978Q3-1996Q3	0.065	3.09**	0.058	2.83**	20.24%

Country		MKT		SMB		adj. R2
		slope	t-stat.	slope	t-stat.	
Australia	1985Q3-1996Q2	0.017	1.02	0.049	3.40**	33.49%
Canada	1978Q3-1996Q2	0.051	3.24**	0.052	2.51**	24.73%
France	1984Q1-1996Q2	0.007	0.65	0.051	3.14**	15.54%
Germany	1987Q4-1996Q3	0.082	2.13**	0.164	2.88**	28.00%
Italy	1988Q1-1996Q3	0.026	2.54**	0.047	3.14**	36.43%
Japan	1981Q2-1996Q3	0.029	1.93*	0.017	1.24	13.18%
Netherlands	1979Q1-1996Q3	0.027	2.14**	0.073	4.79**	26.84%
Switzerland	1987Q1-1995Q4	0.005	0.30	-0.054	-1.85*	25.50%
United Kingdom	1978Q3-1996Q3	0.055	2.98**	0.057	3.90**	32.35%
United States	1978Q3-1996Q3	0.043	2.41**	0.017	0.63	11.49%

Country		MKT		WML		adj. R2
		slope	t-stat.	slope	t-stat.	
Australia	1985Q3-1996Q2	0.026	1.29	0.009	0.38	5.57%
Canada	1978Q3-1996Q2	0.049	2.81**	0.014	1.00	16.38%
France	1984Q1-1996Q2	0.012	0.91	-0.009	-0.44	-1.45%
Germany	1987Q4-1996Q3	0.029	0.76	0.122	1.11	14.46%
Italy	1988Q1-1996Q3	0.032	2.85**	0.002	0.17	23.07%
Japan	1981Q2-1996Q3	0.025	1.50	0.037	1.47	15.24%
Netherlands	1979Q1-1996Q3	0.031	2.05**	-0.041	-2.25**	16.67%
Switzerland	1987Q1-1995Q4	0.018	0.88	-0.013	-0.50	-1.00%
United Kingdom	1978Q3-1996Q3	0.072	3.57**	-0.027	-0.65	19.11%
United States	1978Q3-1996Q3	0.041	1.95*	0.022	0.49	11.07%

* Significant at the 10% Level.

* Significant at the 10% level.

** Significant at the 5% level.

T-statistics are adjusted for heteroscedasticity and serial correlation using the Newey West (1987) estimator. The parameter q was set equal to three. MKT is the excess return on the local market index. In the regression notation, "FactorRet" stands for HML, SMB, and WML. HML is the return on a portfolio that is long on high book-to-market stocks and short on low book-to-market stocks, holding the size and momentum characteristics of the portfolio constant. SMB is the return on a portfolio that is long on small capitalization stocks and short on large capitalization stocks, holding the book-to-market and momentum characteristics of the portfolio constant. WML is the return on a portfolio that is long on the best performing stocks of the past year ("winners") and short on the worst performing stocks of the past year ("losers"), holding the book-to-market and size effects of the portfolio constant. GDP growth is calculated as the continuously compounded growth rate in a country's Gross Domestic Product, which is seasonally adjusted. For Japan we use the seasonally adjusted Gross National Product. All returns are annualized and continuously compounded.

Table 5: Predicting Annual GDP Growth Rates Conditional on Information about the Return on the Market and HML, SMB, and WML Trading Strategies :
Multivariate Regressions

$$\text{GDPgrowth}(t,t+4) = a + b*\text{MKT}(t-4,t) + c*\text{HML}(t-4,t) + d*\text{SMB}(t-4,t) + e(t,t+4)$$

Country	Time Period	MKT		HML		SMB		adj. R2
		slope	t-stat.	slope	t-stat.	slope	t-stat.	
Australia	1985Q3-1996Q2	0.015	0.94	-0.008	-0.41	0.054	2.43**	32.10%
Canada	1978Q3-1996Q2	0.048	3.18**	-0.028	-1.30	0.054	2.49**	26.11%
France	1984Q1-1996Q2	-0.007	-0.93	0.070	4.49**	0.015	1.01	41.40%
Germany	1987Q4-1996Q3	0.081	1.88*	0.008	0.10	0.016	2.77**	25.78%
Italy	1988Q1-1996Q3	0.020	1.82*	0.035	1.33	0.023	1.30	38.76%
Japan	1981Q2-1996Q3	0.030	2.01**	0.014	0.86	0.019	1.40	12.73%
Netherlands	1979Q1-1996Q3	0.030	2.90**	0.010	0.55	0.068	4.06**	26.39%
Switzerland	1987Q1-1995Q4	0.006	0.32	-0.002	-0.09	-0.054	-1.62	23.23%
United Kingdom	1978Q3-1996Q3	0.057	3.42**	0.045	1.67*	0.043	2.21**	36.53%
United States	1978Q3-1996Q3	0.059	3.20**	0.062	2.85**	0.026	0.99	21.30%

* Significant at the 10% Level.

** Significant at the 5% Level.

T-statistics are adjusted for heteroscedasticity and serial correlation using the Newey West (1987) estimator. The parameter q was set equal to three. MKT is the excess return on the local market index. HML is the return on a portfolio that is long on high book-to-market stocks and short on low book-to-market stocks, holding the size and momentum characteristics of the portfolio constant. SMB is the return on a portfolio that is long on small capitalization stocks and short on large capitalization stocks, holding the book-to-market and momentum characteristics of the portfolio constant. WML is the return on a portfolio that is long on the best performing stocks of the past year (“winners”) and short on the worst performing stocks of the past year (“losers”), holding the book-to-market and size effects of the portfolio constant. GDP growth is calculated as the continuously compounded growth rate in a country’s Gross Domestic Product, which is seasonally adjusted. For Japan we use the seasonally adjusted Gross National Product. All returns are annualized and continuously compounded.

Table 6A: The Ability of Stock Market Factors to Predict Annual GDP Growth in the Presence of Business Cycle Variables

$$\text{GDPgrowth}(t,t+4) = a + b \cdot \text{MKT}(t-4,t) + c \cdot \text{FactorRet}(t-4,t) + d \cdot \text{TB}(t) + f \cdot \text{DY}(t) + g \cdot \text{TERM}(t) + h \cdot \text{IDPgrowth}(t-4,t) + e(t,t+4)$$

Country	Time Period	MKT		HML		TB		DY		TERM		IDF growth		adj. R2
		slope	t-stat.	slope	t-stat.	slope	t-stat.	slope	t-stat.	slope	t-stat.	slope	t-stat.	
Australia	1985Q3-1995Q2	-0.048	-5.73**	-0.001	-0.12	-0.247	1.55	-3.168	-8.41**	-0.590	-2.39**	-0.105	-2.14**	72.5%
Canada	1978Q3-1996Q2	0.046	3.91**	-0.035	-2.01**	-0.042	-0.24	0.530	0.86	0.826	3.70**	0.043	0.70	51.8%
France	1984Q1-1996Q2	-0.008	-1.02	0.076	8.37**	-0.143	-1.02	-0.276	-0.99	0.086	0.40	-0.049	-0.93	46.7%
Germany	1987Q4-1996Q3	0.133	2.30**	0.101	1.21	2.639	2.15**	5.234	1.48	3.444	2.19**	0.108	0.73	37.0%
Italy	1988Q1-1996Q3	0.018	2.93**	0.028	2.14**	-0.428	-2.43**	0.067	0.13	0.043	0.29	-0.021	-0.54	72.3%
Japan	1981Q2-1996Q3	0.022	1.81*	0.019	1.82*	0.197	0.72	-0.950	-0.94	0.205	0.59	0.245	4.21**	48.9%
Netherlands	1979Q1-1996Q3	0.013	0.13	0.017	1.14	-0.805	-3.22**	0.216	0.76	-0.619	-2.16**	0.038	0.94	43.6%
Switzerland	1987Q1-1995Q4	0.001	0.03	-0.035	-1.05	1.007	2.49**	-3.102	-2.42**	0.670	1.17	0.140	1.54	41.0%
United Kingdom	1978Q3-1996Q3	0.027	1.47	0.063	3.16**	0.038	0.22	-1.030	-1.47	0.678	3.19**	-0.067	-0.79	59.1%
United States	1978Q3-1996Q3	0.062	2.53**	-0.004	-0.16	-0.002	-0.01	0.689	1.48	0.671	3.50**	0.044	0.70	40.6%

Country	Time Period	MKT		SMB		TB		DY		TERM		IDF growth		adj. R2
		slope	t-stat.	slope	t-stat.	slope	t-stat.	slope	t-stat.	slope	t-stat.	slope	t-stat.	
Australia	1985Q3-1995Q2	-0.047	-6.42**	0.011	0.90	-0.139	-0.84	-3.048	-8.91**	-0.433	-1.64	-0.090	-1.85*	73.0%
Canada	1978Q3-1996Q2	0.053	4.16**	-0.002	-0.11	-0.107	-0.45	0.768	1.21	0.768	3.09**	0.015	0.24	48.4%
France	1984Q1-1996Q2	0.000	0.02	0.056	3.39**	0.044	0.26	-1.061	-2.09**	0.285	1.05	-0.153	-2.34**	29.5%
Germany	1987Q4-1996Q3	0.186	2.64**	0.191	2.82**	2.247	1.89*	5.203	1.50	3.948	2.44**	-0.225	0.17	47.4%
Italy	1988Q1-1996Q3	0.022	2.68**	0.040	2.42**	-0.234	-1.44	-0.626	-1.10	0.120	0.92	-0.079	-1.27	74.0%
Japan	1981Q2-1996Q3	0.023	2.06**	-0.019	-1.34	0.233	0.82	-1.189	-1.07	0.130	0.35	0.254	4.38**	49.2%
Netherlands	1979Q1-1996Q3	0.012	1.45	0.052	4.63**	-0.740	-3.30**	0.200	0.78	-0.550	-2.27**	-0.016	-0.40	50.5%
Switzerland	1987Q1-1995Q4	-0.013	-0.80	-0.027	-0.86	0.975	2.48**	-3.079	-2.45**	0.765	1.49	0.103	1.36	42.5%
United Kingdom	1978Q3-1996Q3	0.018	0.84	0.006	0.31	0.278	1.38	-1.756	-1.97**	0.813	2.89**	-0.008	-0.08	51.7%
United States	1978Q3-1996Q3	0.051	2.96**	0.039	1.76*	0.079	0.42	0.513	1.30	0.775	3.91**	0.012	0.19	45.1%

Country	Time Period	MKT		WML		TB		DY		TERM		IDF growth		adj. R2
		slope	t-stat.	slope	t-stat.	slope	t-stat.	slope	t-stat.	slope	t-stat.	slope	t-stat.	
Australia	1985Q3-1995Q2	-0.050	-4.87**	0.003	0.26	-0.227	-1.93*	-3.190	-7.87**	-0.567	-2.71**	-0.1018	-2.05**	72.6%
Canada	1978Q3-1996Q2	0.051	4.36**	0.012	0.96	-0.077	-0.39	0.833	1.33	0.784	3.15**	0.032	0.47	49.2%
France	1984Q1-1996Q2	0.002	0.15	-0.006	-0.38	0.124	0.66	-1.164	-2.57**	0.353	1.30	-0.005	-0.08	15.2%
Germany	1987Q4-1996Q3	0.091	1.69*	0.101	1.53	1.563	1.57	4.306	1.34	1.667	1.32	0.277	1.84*	39.5%
Italy	1988Q1-1996Q3	0.024	1.33	-0.016	-0.53	-0.462	-1.05	0.048	0.07	-0.004	-0.01	0.009	0.24	70.6%
Japan	1981Q2-1996Q3	0.020	1.73*	0.012	0.56	0.163	0.56	-0.796	-0.75	0.117	0.32	0.236	3.97**	47.5%
Netherlands	1979Q1-1996Q3	0.011	1.31	-0.037	-2.82**	-0.824	-4.28**	0.250	0.99	-0.585	-2.29**	0.031	0.72	50.6%
Switzerland	1987Q1-1995Q4	0.001	0.11	-0.028	-1.75*	1.290	4.14**	-2.964	-2.50**	1.147	3.51**	0.096	1.25	41.3%
United Kingdom	1978Q3-1996Q3	0.001	0.04	0.057	1.81*	0.455	1.93*	-2.371	-2.52**	1.050	3.73**	0.003	0.04	54.8%
United States	1978Q3-1996Q3	0.036	1.82*	0.086	3.38**	0.108	0.64	0.502	1.29	0.822	4.68**	0.090	1.60	47.4%

* Significant at the 10% Level.

** Significant at the 5% Level.

T-statistics are adjusted for heteroscedasticity and serial correlation using the Newey West (1987) estimator. The parameter q was set equal to three. MKT is the excess return on the local market index. In the regression notation, "FactorRet" stands for HML, SMB, and WML. HML is the return on a portfolio that is long on high book-to-market stocks and short on low book-to-market stocks, holding the size and momentum characteristics of the portfolio constant. SMB is the return on a portfolio that is long on small capitalization stocks and short on large capitalization stocks, holding the book-to-market and momentum characteristics of the portfolio constant. WML is the return on a portfolio that is long on the best performing stocks of the past year ("winners") and short on the worst performing stocks of the past year ("losers"), holding the book-to-market and size effects of the portfolio constant. TB denotes the annual Treasury Bill yield. DY is the dividend yield on a country's market capitalization weighted index. TERM stands for the premium on a ten-year government bond minus the TB rate. IDP growth is the growth in a country's industrial production. GDP growth is calculated as the continuously compounded growth rate in a country's Gross Domestic Product, which is seasonally adjusted. For Japan we use the seasonally adjusted Gross National Product. All returns are annualized and continuously compounded.

Table 6B: The Ability of Stock Market Factors to Predict Annual GDP Growth in the Presence of Business Cycle Variables

$$\text{GDPgrowth}(t,t+4) = a + b \cdot \text{MKT}(t-4,t) + c \cdot \text{HML}(t-4,t) + d \cdot \text{SMB}(t-4,t) + e \cdot \text{TB}(t) + g \cdot \text{DY}(t) + h \cdot \text{TERM}(t) + i \cdot \text{IDPgrowth}(t-4,t) + e(t,t+4)$$

Country	Time Period	MKT		HML		SMB		TB		DY		TERM		IDP growth		adj. R2
		slope	t-stat.	slope	t-stat.	slope	t-stat.	slope	t-stat.	slope	t-stat.	slope	t-stat.	slope	t-stat.	
Australia	1985Q3-1995Q2	-0.048	-6.22**	-0.006	-0.42	0.013	1.00	-0.165	-0.93	-3.037	-8.91**	-0.481	-1.77*	-0.091	-1.86*	72.1%
Canada	1978Q3-1996Q2	0.047	3.77**	-0.036	-2.00**	0.004	0.18	-0.030	-0.14	0.519	0.82	0.824	3.72**	0.044	0.72	51.0%
France	1984Q1-1996Q2	-0.007	-0.87	0.066	6.32**	0.028	1.77*	-0.141	-1.10	-0.335	-1.15	0.097	0.45	-0.118	-1.97**	48.4%
Germany	1987Q4-1996Q3	0.174	2.58**	0.089	1.10	0.187	2.77**	2.677	2.13**	4.520	1.32	4.430	2.69**	-0.280	-1.94**	47.7%
Italy	1988Q1-1996Q3	0.024	2.85**	-0.006	-0.20	0.045	1.48	-0.196	-0.73	-0.734	-0.80	0.142	0.74	-0.084	-1.15	72.2%
Japan	1981Q2-1996Q3	0.022	2.05**	0.016	1.25	-0.017	-1.13	0.225	0.82	-1.261	-1.16	0.174	0.46	0.258	4.58**	49.5%
Netherlands	1979Q1-1996Q3	0.013	1.54	0.005	0.37	0.049	4.58**	-0.735	-3.22**	0.196	0.75	-0.552	-2.36**	-0.012	-0.31	49.9%
Switzerland	1987Q1-1995Q4	-0.009	-0.46	-0.018	-0.74	-0.022	-0.68	0.914	2.03**	-3.125	-2.50**	0.632	1.02	0.128	1.86*	41.1%
United Kingdom	1978Q3-1996Q3	0.027	1.48	0.064	3.00**	-0.004	-0.22	0.043	0.25	-1.057	-1.46	0.695	3.14**	-0.063	-0.74	58.5%
United States	1978Q3-1996Q3	0.051	2.72**	0.000	0.01	0.039	1.71*	0.079	0.43	0.513	1.30	0.774	4.30**	0.012	0.18	44.3%

* Significant at the 10% Level.

** Significant at the 5% Level.

T-statistics are adjusted for heteroscedasticity and serial correlation using the Newey West (1987) estimator. The parameter q was set equal to three. MKT is the excess return on the local market index. HML is the return on a portfolio that is long on high book-to-market stocks and short on low book-to-market stocks, holding the size and momentum characteristics of the portfolio constant. SMB is the return on a portfolio that is long on small capitalization stocks and short on large capitalization stocks, holding the book-to-market and momentum characteristics of the portfolio constant. WML is the return on a portfolio that is long on the best performing stocks of the past year (“winners”) and short on the worst performing stocks of the past year (“losers”), holding the book-to-market and size effects of the portfolio constant. TB denotes the annual Treasury Bill yield. DY is the dividend yield on a country’s market capitalization weighted index. TERM stands for the premium on a ten-year government bond minus the TB rate. IDP growth is the growth in a country’s industrial production. GDP growth is calculated as the continuously compounded growth rate in a country’s Gross Domestic Product, which is seasonally adjusted. For Japan we use the seasonally adjusted Gross National Product. All returns are annualized and continuously compounded.

Table 7: Analysis of Portfolio Holdings. Average Percentage of Securities in the Extreme Tritle Portfolios that Remain the Same Over Time – Annual Rebalancing

Years		One year			Two Years			Three Years			Four Years			Five Years			Six Years			Seven Years			Eight Years			Nine Years			Ten Years		
		Same	Change	Absent	Same	Change	Absent	Same	Change	Absent	Same	Change	Absent	Same	Change	Absent	Same	Change	Absent	Same	Change	Absent	Same	Change	Absent	Same	Change	Absent	Same	Change	Absent
Australia	High B/M	71%	16%	13%	59%	19%	21%	50%	22%	29%	45%	19%	36%	37%	20%	43%	29%	22%	48%	24%	24%	52%	20%	25%	55%	21%	27%	52%	18%	33%	49%
	Low B/M	73%	23%	4%	64%	30%	5%	61%	34%	5%	56%	37%	8%	52%	38%	9%	50%	42%	8%	48%	46%	6%	48%	45%	7%	48%	43%	9%	45%	45%	10%
	Small	71%	20%	8%	61%	27%	12%	54%	32%	15%	49%	34%	17%	47%	37%	16%	49%	35%	16%	48%	38%	14%	44%	42%	14%	44%	42%	14%	47%	39%	13%
	Big	81%	15%	4%	78%	15%	7%	78%	11%	11%	76%	10%	15%	73%	9%	18%	69%	9%	21%	70%	7%	24%	71%	5%	24%	72%	6%	23%	70%	8%	22%
	Winners	33%	63%	4%	29%	63%	8%	31%	60%	9%	27%	61%	12%	25%	59%	15%	25%	55%	20%	23%	55%	22%	23%	56%	22%	28%	49%	23%	20%	54%	26%
Canada	Losers	33%	59%	8%	29%	59%	12%	26%	59%	15%	27%	55%	18%	23%	57%	20%	24%	55%	21%	28%	50%	22%	25%	51%	24%	42%	35%	23%	18%	62%	20%
	High B/M	74%	23%	4%	64%	29%	7%	59%	31%	11%	55%	31%	14%	50%	32%	18%	47%	32%	21%	44%	32%	24%	41%	32%	27%	38%	31%	31%	36%	30%	34%
	Low B/M	75%	20%	4%	65%	28%	7%	58%	32%	10%	53%	35%	13%	48%	37%	15%	42%	39%	19%	37%	41%	22%	33%	41%	26%	30%	42%	29%	29%	39%	32%
	Small	78%	17%	6%	69%	22%	9%	62%	27%	12%	55%	30%	16%	49%	32%	18%	46%	33%	21%	44%	32%	24%	41%	31%	27%	38%	33%	30%	35%	33%	32%
	Big	88%	9%	3%	84%	11%	5%	82%	10%	7%	80%	10%	10%	79%	9%	12%	77%	9%	14%	75%	8%	17%	74%	7%	20%	72%	5%	23%	70%	4%	25%
France	Winners	31%	65%	4%	29%	65%	6%	33%	58%	9%	28%	60%	12%	26%	59%	15%	27%	54%	18%	25%	54%	21%	25%	50%	25%	23%	50%	27%	22%	47%	31%
	Losers	32%	64%	4%	32%	60%	8%	33%	56%	11%	29%	57%	14%	28%	55%	17%	28%	52%	19%	26%	53%	21%	28%	48%	25%	21%	50%	29%	21%	46%	33%
	High B/M	77%	19%	3%	66%	28%	6%	58%	32%	9%	50%	37%	13%	45%	39%	16%	40%	40%	20%	36%	41%	23%	32%	46%	22%	29%	46%	26%	30%	42%	28%
	Low B/M	73%	21%	6%	64%	28%	8%	58%	32%	10%	53%	34%	13%	50%	35%	15%	45%	38%	17%	46%	34%	20%	46%	33%	21%	44%	32%	24%	40%	34%	26%
	Small	76%	19%	6%	67%	25%	8%	59%	30%	11%	49%	36%	14%	39%	44%	17%	28%	52%	20%	24%	54%	23%	21%	54%	25%	21%	49%	30%	17%	52%	31%
Germany	Big	84%	12%	4%	82%	12%	6%	81%	10%	9%	81%	9%	11%	79%	8%	13%	80%	5%	16%	78%	4%	18%	79%	4%	17%	77%	5%	17%	72%	6%	22%
	Winners	36%	60%	4%	30%	63%	7%	28%	61%	10%	30%	56%	14%	28%	57%	15%	30%	53%	17%	25%	56%	20%	29%	49%	22%	23%	52%	25%	23%	52%	24%
	Losers	36%	58%	6%	32%	60%	8%	33%	58%	9%	30%	58%	12%	28%	56%	16%	28%	55%	17%	28%	54%	18%	34%	47%	19%	31%	51%	18%	25%	56%	19%
	High B/M	80%	17%	3%	71%	24%	4%	67%	28%	5%	64%	30%	6%	60%	32%	8%	51%	38%	11%	47%	41%	12%	50%	36%	14%	50%	33%	17%			
	Low B/M	70%	24%	6%	59%	33%	8%	54%	38%	8%	51%	41%	8%	48%	44%	7%	47%	46%	7%	40%	58%	1%	36%	58%	6%	23%	62%	15%			
Italy	Small	72%	21%	8%	60%	30%	9%	47%	43%	10%	37%	52%	11%	32%	61%	7%	26%	68%	7%	18%	78%	4%	16%	81%	3%	25%	67%	8%			
	Big	91%	7%	2%	92%	5%	3%	92%	5%	4%	91%	5%	4%	91%	3%	6%	92%	2%	5%	94%	1%	5%	93%	1%	6%	92%	0%	8%			
	Winners	39%	57%	4%	27%	68%	5%	28%	66%	6%	30%	63%	7%	29%	63%	8%	24%	67%	9%	33%	59%	8%	29%	66%	5%	38%	46%	15%			
	Losers	37%	58%	4%	31%	64%	5%	32%	62%	6%	32%	62%	6%	30%	63%	7%	23%	69%	8%	32%	59%	9%	22%	62%	16%	25%	58%	17%			
	High B/M	81%	16%	2%	73%	23%	4%	61%	32%	6%	54%	40%	6%	53%	41%	7%	46%	48%	6%	40%	48%	12%	45%	52%	3%	45%	55%	0%			
Japan	Low B/M	79%	18%	3%	69%	27%	4%	59%	35%	6%	52%	41%	7%	48%	47%	6%	41%	54%	6%	37%	61%	2%	38%	63%	0%	33%	67%	0%			
	Small	82%	16%	2%	81%	17%	2%	77%	21%	3%	77%	20%	3%	75%	21%	3%	78%	18%	4%	74%	23%	3%	75%	22%	3%	64%	36%	0%			
	Big	89%	10%	1%	88%	10%	2%	85%	12%	2%	86%	12%	2%	89%	9%	3%	87%	10%	3%	87%	9%	4%	88%	13%	0%	83%	17%	0%			
	Winners	35%	63%	2%	34%	63%	3%	35%	61%	4%	35%	60%	5%	31%	66%	4%	41%	50%	9%	30%	60%	10%	35%	60%	6%	50%	50%	0%			
	Losers	35%	62%	3%	37%	59%	4%	33%	62%	5%	28%	67%	5%	28%	69%	3%	35%	62%	3%	34%	64%	2%	20%	80%	0%	27%	73%	0%			
Netherl.	High B/M	75%	25%	1%	65%	34%	1%	59%	39%	1%	58%	41%	1%	56%	42%	1%	54%	45%	1%	51%	49%	0%	50%	50%	0%	53%	47%	0%	56%	43%	0%
	Low B/M	78%	20%	1%	69%	30%	2%	61%	36%	2%	55%	42%	3%	50%	46%	3%	48%	49%	3%	44%	54%	2%	41%	58%	2%	40%	59%	1%	38%	60%	1%
	Small	83%	16%	1%	76%	23%	2%	68%	29%	3%	66%	31%	3%	63%	34%	4%	60%	37%	3%	58%	40%	2%	52%	46%	2%	48%	50%	1%	44%	55%	1%
	Big	92%	8%	0%	91%	9%	0%	92%	7%	0%	92%	7%	1%	93%	6%	1%	94%	6%	0%	94%	5%	0%	95%	5%	0%	95%	4%	0%	96%	4%	0%
	Winners	30%	70%	0%	30%	69%	1%	27%	71%	1%	30%	69%	1%	30%	69%	1%	28%	71%	2%	30%	68%	1%	32%	67%	1%	27%	72%	1%	29%	71%	1%
Switzerl.	Losers	32%	67%	1%	33%	66%	1%	37%	61%	1%	36%	62%	1%	34%	63%	2%	32%	66%	2%	34%	66%	1%	35%	64%	1%	32%	67%	1%	30%	69%	1%
	High B/M	81%	12%	8%	72%	13%	15%	66%	11%	23%	57%	11%	32%	48%	11%	41%	40%	11%	50%	35%	8%	57%	31%	6%	63%	27%	4%	68%	24%	3%	73%
	Low B/M	79%	20%	1%	71%	28%	1%	66%	31%	3%	60%	37%	3%	57%	39%	4%	50%	44%	6%	46%	47%	7%	46%	47%	7%	47%	45%	8%	46%	44%	10%
	Small	77%	20%	4%	68%	27%	5%	61%	32%	7%	53%	38%	9%	49%	40%	11%	46%	38%	15%	42%	40%	18%	43%	36%	22%	40%	36%	24%	37%	36%	27%
	Big	84%	13%	3%	79%	15%	6%	78%	12%	10%	75%	13%	13%	70%	14%	16%	65%	15%	20%	63%	15%	22%	59%	15%	26%	58%	14%	28%	59%	12%	30%
U.K.	Winners	37%	61%	3%	32%	62%	6%	29%	62%	9%	28%	60%	13%	25%	60%	15%	31%	51%	18%	26%	53%	21%	29%	48%	23%	22%	50%	28%	24%	48%	29%
	Losers	34%	61%	5%	27%	66%	7%	26%	63%	11%	25%	61%	14%	27%	56%	17%	28%	49%	23%	26%	48%	26%	21%	46%	33%	24%	41%	35%	21%	41%	38%
	High B/M	78%	20%	2%	70%	28%	3%	60%	36%	3%	51%	45%	5%	46%	46%	8%	32%	61%	7%	22%	73%	5%	10%	88%	2%	6%	94%	0%			
	Low B/M	78%	18%	5%	68%	24%	7%	61%	30%	9%	59%	27%	14%	49%	32%	19%	45%	34%	20%	40%	37%	24%	38%	32%	29%	22%	39%	39%			
	Small	74%	20%	6%	66%	26%	8%	60%	30%	10%	60%	30%	10%	57%	28%	16%	52%	30%	18%	50%	35%	16%	34%	44%	22%	35%	35%	29%			
U.S.	Big	86%	12%	2%	84%	12%	4%	83%	12%	5%	79%	13%	7%	76%	11%	13%	75%	12%	13%	77%	11%	13%	73%	15%	12%	78%	11%	11%			
	Winners	41%	57%	2%	34%	61%	5%	32%	62%	6%	25%	65%	10%	29%	52%	19%	27%	56%	17%	25%	58%	17%	25%	50%	26%	11%	67%	22%			
	Losers	37%	58%	5%	32%	61%	7%	28%	66%	6%	32%	59%	10%	33%	57%	10%	33%	52%	15%	27%	57%	16%	17%	66%	16%	18%	53%	29%			
	High B/M	74%	17%	9%	62%	21%	16%	54%	23%	24%	47%	24%	30%	41																	

Table 8: Percentage Concentration by Industry in Extreme Portfolios – Annual Rebalancing

		Australia		Canada		France		Germany		Italy		Japan		Netherlands		Switzerland		United Kingdom		United States	
All	Consumer Goods	13%		11%		18%		21%		13%		15%		21%		24%		11%		14%	
	Financials	3%		9%		13%		9%		22%		0%		14%		15%		9%		16%	
	General Industrials	32%		23%		35%		46%		40%		56%		33%		41%		30%		29%	
	Resources	36%		31%		4%		1%		0%		4%		2%		0%		3%		7%	
	Services	16%		24%		29%		17%		23%		22%		29%		14%		27%		26%	
	Utilities	0%		3%		1%		6%		2%		2%		0%		4%		1%		7%	
	Others	0%		0%		0%		0%		1%		0%		0%		1%		19%		0%	
B/M		High	Low	High	Low	High	Low	High	Low	High	Low	High	Low	High	Low	High	Low	High	Low	High	Low
	Consumer Goods	13%	13%	14%	9%	12%	22%	19%	25%	10%	12%	15%	15%	14%	28%	23%	36%	9%	12%	10%	22%
	Financials	3%	1%	15%	4%	17%	6%	10%	9%	19%	28%	1%	0%	24%	6%	9%	14%	13%	8%	28%	6%
	General Industrials	41%	16%	27%	18%	44%	26%	43%	45%	41%	34%	55%	56%	35%	28%	53%	30%	24%	30%	24%	30%
	Resources	25%	52%	23%	40%	6%	1%	1%	1%	0%	0%	5%	4%	0%	4%	0%	0%	1%	3%	7%	7%
	Services	18%	19%	19%	27%	21%	42%	17%	17%	29%	24%	23%	24%	26%	35%	15%	14%	19%	36%	21%	34%
	Utilities	0%	0%	2%	2%	0%	3%	11%	2%	0%	3%	1%	1%	0%	0%	1%	4%	1%	0%	10%	1%
	Others	0%	0%	0%	0%	0%	0%	0%	1%	1%	0%	0%	0%	0%	0%	0%	1%	31%	10%	1%	0%
ME		Small	Big	Small	Big	Small	Big	Small	Big	Small	Big	Small	Big	Small	Big	Small	Big	Small	Big	Small	Big
	Consumer Goods	11%	15%	15%	9%	14%	23%	33%	9%	24%	8%	15%	15%	34%	14%	27%	29%	13%	11%	15%	14%
	Financials	2%	4%	4%	14%	5%	15%	0%	19%	8%	37%	0%	1%	4%	26%	5%	25%	5%	15%	11%	18%
	General Industrials	29%	32%	22%	23%	42%	29%	44%	45%	48%	21%	61%	53%	36%	27%	41%	33%	29%	28%	32%	25%
	Resources	42%	32%	33%	31%	1%	8%	1%	1%	0%	0%	4%	4%	0%	6%	0%	0%	3%	3%	7%	8%
	Services	16%	17%	26%	17%	36%	23%	19%	15%	18%	29%	19%	24%	26%	27%	23%	8%	29%	24%	32%	22%
	Utilities	0%	0%	1%	6%	0%	3%	1%	11%	0%	5%	1%	3%	0%	0%	3%	3%	0%	2%	1%	13%
	Others	0%	0%	0%	0%	1%	0%	0%	0%	2%	0%	0%	0%	0%	0%	0%	2%	21%	16%	1%	0%
Past Returns		Losers	Winners	Losers	Winners	Losers	Winners	Losers	Winners	Losers	Winners	Losers	Winners	Losers	Winners	Losers	Winners	Losers	Winners	Losers	Winners
	Consumer Goods	11%	11%	10%	11%	18%	18%	21%	20%	12%	13%	15%	14%	21%	21%	23%	23%	12%	11%	14%	15%
	Financials	2%	3%	7%	10%	13%	10%	11%	6%	26%	18%	0%	1%	19%	10%	20%	12%	10%	9%	14%	17%
	General Industrials	32%	28%	23%	24%	37%	36%	44%	51%	41%	40%	55%	58%	29%	34%	39%	48%	30%	29%	29%	29%
	Resources	40%	42%	34%	31%	5%	3%	1%	1%	0%	0%	4%	4%	1%	3%	0%	0%	3%	3%	9%	7%
	Services	14%	15%	23%	22%	25%	33%	18%	14%	20%	26%	24%	21%	30%	32%	15%	12%	25%	30%	26%	27%
	Utilities	0%	0%	3%	3%	1%	1%	4%	7%	1%	2%	1%	2%	0%	0%	4%	5%	1%	1%	8%	5%
	Others	0%	0%	0%	0%	0%	1%	1%	0%	0%	1%	0%	0%	0%	0%	0%	1%	19%	17%	0%	0%

“High B/M” (book-to-market) and “Low B/M” are extreme portfolios based on tritile sorts of book-to-market ratios. “Small” and “Big” are extreme tritile portfolios of securities sorted according to market capitalization, after stocks have been sorted according to their B/M. “Winners” and “Losers” are extreme tritile portfolios of securities sorted on the basis of their past year’s returns, after securities have been first sorted on the basis of their book-to-market and market capitalization. Industry classification is based on Datastream’s Level 3 industrial codes. “Others” include suspended equities and missing observations.

Appendix 1
Universe of Country Equity Price Data

Country	Total Number of Stocks	Surviving Stocks	Defunct Stocks
Australia	332	193	139
Canada	812	416	396
France	467	308	159
Germany	612	545	67
Italy	173	132	41
Japan	1,152	1,105	47
Netherlands	228	106	122
Switzerland	196	143	53
United Kingdom	2,131	862	1,269
United States	3,650	1,830	1,820
Total	9,753	5,640	4,113

Equity data are obtained from Datastream International. The “Surviving Stocks” are from the “Country List” of Datastream. The “Defunct Stocks” are from the “Deadlist”.

Appendix 2: Comparison of Constructed Market Capitalization Weighted Indexes with the Morgan Stanley Capital International Indexes

Country	Constructed Mkt. Cap. Weighted Index				MSCI Mkt. Cap. Weighted Index				corr.	t-stat.	p-val.	F-stat.	p.val
	mean	Std.	skewness	kurtosis	mean	std.	skewness	kurtosis					
Australia	17.87%	21.29%	-1.47	10.89	17.39%	20.66%	-1.48	10.65	0.98	0.45	0.65	1.06	0.32
Canada	14.32%	15.80%	-0.81	3.80	14.07%	16.69%	-0.53	3.01	0.98	0.31	0.76	0.90	0.80
France	18.95%	19.85%	-0.19	1.86	18.59%	20.78%	-0.14	1.40	0.98	0.40	0.69	0.91	0.76
Germany	11.64%	12.76%	-0.66	3.38	12.76%	17.79%	-0.59	2.94	0.98	-1.31	0.19	0.79**	0.96
Italy	20.74%	26.34%	0.71	1.73	19.75%	25.94%	0.63	1.17	0.98	0.75	0.45	1.03	0.41
Japan	10.25%	17.77%	-0.18	2.83	10.61%	18.64%	-0.08	1.77	0.92	-0.76	0.45	0.84*	0.91
Netherlands	17.34%	15.33%	-0.67	4.26	18.17%	16.11%	-0.43	3.43	0.97	-1.03	0.30	0.91	0.78
Switzerland	12.11%	14.12%	-1.02	5.64	13.67%	15.44%	-0.70	3.92	0.98	-2.04**	0.04	0.83*	0.91
United Kingdom	18.75%	16.95%	-0.87	3.92	18.11%	16.18%	-0.81	3.59	0.98	0.91	0.36	0.96	0.62
United States	15.93%	14.67%	-0.71	3.62	15.50%	16.61%	-0.56	3.40	0.98	0.56	0.57	1.04	0.39

*** significant at the 5% level.

* significant at the 10% level.

All returns are annualized and in local currency. “Corr” stands for correlation between the returns of the constructed indexes and the MSCI indexes. T-stat and F-stat are two tailed tests of the null hypotheses of equality of means and variances, respectively. The data span the period from August 1976 to May 1997 and they are from Datastream.

Appendix 3: Number of Securities Included in the July Portfolios of the Annual Portfolio Sorts

Year(July)	Australia	Canada	France	Germany	Italy	Japan	Netherlands	Switzerland	U.K.	U.S.
1978		140							1,079	1,546
1979		145					28		1,129	1,593
1980		150					36		1,142	1,637
1981		157				67	40		1,176	1,661
1982		170				85	42		1,186	1,734
1983		213				85	44		1,168	1,788
1984		224	100			656	41		1,173	1,811
1985	71	240	105			757	42		1,147	1,854
1986	88	248	111			798	42		1,099	1,844
1987	99	261	125	38	35	814	48	53	1,041	1,877
1988	102	296	129	134	48	823	48	67	1,029	1,911
1989	157	293	155	139	85	994	50	109	979	1,839
1990	160	317	281	224	89	1,049	53	128	918	1,763
1991	169	329	284	342	97	1,066	49	133	893	1,699
1992	175	356	292	379	103	1,072	52	132	849	1,654
1993	169	373	301	401	109	1,070	56	133	822	1,696
1994	182	425	292	409	111	1,074	92	142	809	1,810
1995	175	414	287	416	120	1,077	99	135	832	1,792
1996	175	382	280	441	122	1,086	100	131	817	1,670
avg.	144	270	211	292	92	786	53	116	1,015	1,746

For inclusion in a portfolio, a security must have positive December-end price-to-book values, June-end market capitalization, and twelve months of returns prior to July. Portfolios are held for one year and are rebalanced at the end of June. The data are from Datastream.

Appendix 4: Regression Results of Portfolio Strategy Returns on Excess Market Returns

Country	Time Period	HML			SMB			WML		
		alpha	beta	adj. R2	alpha	beta	adj. R2	alpha	beta	adj. R2
Australia	1984M10-1997M5	0.010 3.20**	-0.190 -3.45**	6.7%	0.007 1.92*	-0.230 -3.49**	6.9%	0.006 1.74*	0.203 3.29**	6.1%
Canada	1977M10-1997M5	0.010 4.77**	-0.170 -3.27**	5.2%	0.007 3.58**	-0.207 -5.19**	9.9%	0.011 4.36**	0.141 2.56**	2.3%
France	1983M4-1997M5	0.010 4.59**	0.111 2.87**	4.1%	0.005 1.89*	0.020 0.45	-0.5%	0.010 4.63**	0.091 2.42**	2.8%
Germany	1987M1-1997M5	0.006 3.03**	0.006 0.18	-0.8%	0.002 0.78	-0.331 -8.39**	29.8%	0.008 3.58**	-0.088 -1.86*	1.9%
Italy	1987M4-1997M5	0.008 2.95**	0.108 2.84**	5.5%	0.001 0.21	-0.033 -0.82	-0.3%	0.005 1.53	-0.138 -2.83**	5.5%
Japan	1980M7-1997M5	0.011 5.79**	-0.130 -3.72**	6.0%	0.005 1.65	0.008 0.13	-0.5%	0.000 0.22	0.030 0.71	-0.2%
Netherlands	1978M4-1997M5	-0.002 -0.01	0.002 0.03	-0.4%	0.000 0.17	-0.065 -1.23	0.2%	0.012 4.74**	-0.030 -0.55	-0.3%
Switzerland	1986M4-1997M5	0.004 1.65	0.183 3.39**	7.3%	-0.001 -0.370	-0.209 -3.95**	9.9%	0.007 2.89**	0.052 1.01	0.0%
United Kingdom	1977M10-1997M5	0.007 5.75**	-0.056 -2.21**	1.6%	0.006 3.00**	-0.324 -8.23**	22.1%	0.013 9.94**	0.051 1.98**	1.2%
United States	1977M10-1997M5	0.010 4.77**	-0.313 -6.33**	14.3%	0.009 3.57**	0.055 0.94	-0.1%	0.011 4.87**	0.240 4.60**	7.9%

* Significant at the 10% Level.

** Significant at the 5% Level.

T-values appear below the coefficient estimates and they are adjusted for heteroskedasticity and serial correlation. HML is the return on a portfolio that is long on high book-to-market stocks and short on low book-to-market stocks, holding the size and momentum characteristics of the portfolio constant. SMB is the return on a portfolio that is long on small capitalization stocks and short on large capitalization stocks, holding the book-to-market and momentum characteristics of the portfolio constant. WML is the return on a portfolio that is long on the best performing stocks of the past year (“winners”) and short on the worst performing stocks of the past year (“losers”), holding the book-to-market and size effects of the portfolio constant. The data are from Datastream.

Appendix 5: Summary Statistics

Variables	Time Period	Obs.	Mean	Std. Dev.	Autocorrelations for Quarterly Lag					Correlations								
					1	2	3	4	8	MKT	HML	SMB	WML	DY	TB	TERM	GDP	IDP
Australia																		
MKT	1977Q1-1997Q1	81	7.57%	22.21%	-0.04	0.15	0.05	-0.10	0.06	1.00								
HML	1984Q4-1997Q1	50	10.62%	14.62%	0.03	-0.03	0.12	0.07	-0.10	-0.08	1.00							
SMB	1984Q4-1997Q1	50	8.32%	19.56%	-0.13	0.15	0.15	-0.10	-0.06	-0.14	0.54	1.00						
WML	1984Q4-1997Q1	50	7.80%	12.72%	0.12	-0.08	-0.10	-0.07	0.07	0.25	-0.09	0.01	1.00					
DY	1977Q1-1997Q2	82	3.92%	0.92%	0.86	0.74	0.59	0.44	0.05	-0.36	-0.09	-0.14	-0.11	1.00				
TB	1977Q1-1997Q3	83	9.91%	3.50%	0.91	0.83	0.74	0.64	0.29	0.13	-0.31	-0.34	-0.21	0.45	1.00			
TERM	1977Q1-1997Q3	83	1.44%	1.80%	0.80	0.65	0.49	0.30	-0.13	-0.25	0.24	0.29	0.11	-0.38	-0.92	1.00		
GDP growth	1977Q1-1997Q1	81	3.04%	2.39%	-0.13	0.12	-0.03	0.13	0.04	0.21	0.06	0.23	0.05	-0.49	-0.12	0.05	1.00	
IDP growth	1977Q1-1994Q2	70	1.48%	6.08%	-0.26	-0.16	0.15	0.08	-0.21	0.14	-0.25	-0.33	-0.02	-0.13	-0.06	0.04	-0.24	1.00
Canada																		
MKT	1977Q1-1997Q1	81	4.90%	15.48%	0.19	0.08	0.07	-0.22	0.07	1.00								
HML	1977Q4-1997Q1	78	10.27%	11.59%	0.18	0.05	-0.07	-0.12	-0.03	-0.13	1.00							
SMB	1977Q4-1997Q1	78	4.90%	11.98%	0.17	0.05	0.09	0.31	0.24	-0.12	0.15	1.00						
WML	1977Q4-1997Q1	78	16.33%	14.92%	0.20	0.11	-0.03	-0.04	-0.17	0.11	-0.28	0.22	1.00					
DY	1977Q1-1997Q2	82	3.14%	0.69%	0.91	0.80	0.67	0.55	0.45	-0.15	-0.16	-0.14	-0.07	1.00				
TB	1977Q1-1997Q3	83	8.75%	3.41%	0.90	0.80	0.72	0.62	0.03	-0.23	0.02	-0.31	-0.18	0.68	1.00			
TERM	1977Q1-1997Q3	83	1.27%	1.79%	0.76	0.59	0.45	0.39	0.07	0.04	-0.01	0.29	0.12	-0.35	-0.75	1.00		
GDP growth	1977Q1-1997Q1	81	2.54%	1.70%	0.54	0.29	0.11	0.00	-0.07	0.22	0.16	0.00	-0.05	-0.29	-0.31	0.29	1.00	
IPD growth	1977Q1-1997Q3	83	2.17%	3.85%	0.42	0.32	0.06	-0.17	-0.03	0.13	0.26	0.04	-0.18	-0.37	-0.28	0.25	0.66	1.00
France																		
MKT	1977Q1-1997Q1	81	9.14%	21.19%	0.04	-0.07	0.04	0.03	-0.03	1.00								
HML	1983Q2-1997Q1	56	13.27%	11.63%	0.13	-0.03	0.33	0.31	0.31	0.36	1.00							
SMB	1983Q2-1997Q1	56	5.51%	10.99%	0.08	0.12	-0.12	0.15	0.00	0.05	0.43	1.00						
WML	1983Q2-1997Q1	56	13.13%	12.13%	0.13	0.01	0.21	0.29	0.04	0.17	-0.11	-0.34	1.00					
DY	1977Q1-1997Q2	82	4.35%	1.57%	0.93	0.86	0.81	0.74	0.50	-0.13	-0.20	-0.16	0.16	1.00				
TB	1977Q1-1997Q3	83	9.37%	3.47%	0.77	0.74	0.65	0.55	0.35	0.08	0.16	-0.06	0.18	0.34	1.00			
TERM	1977Q1-1997Q3	83	0.42%	1.85%	0.38	0.46	0.27	0.13	-0.04	-0.13	-0.05	0.13	-0.09	-0.03	-0.74	1.00		
GDP growth	1977Q1-1997Q2	82	2.01%	1.24%	0.06	0.28	0.02	-0.04	-0.19	0.03	0.05	0.20	0.03	-0.31	-0.27	0.34	1.00	
IDP growth	1977Q1-1997Q3	83	1.00%	2.65%	-0.05	0.19	0.03	-0.18	0.06	0.00	0.04	0.33	-0.04	-0.15	-0.24	0.39	0.33	1.00
Germany																		
MKT	1977Q1-1997Q1	81	5.74%	16.89%	0.11	-0.05	0.17	-0.05	-0.13	1.00								
HML	1987Q1-1997Q1	41	5.96%	7.06%	-0.15	0.23	-0.09	0.07	0.02	0.11	1.00							
SMB	1987Q1-1997Q1	41	0.65%	11.41%	0.19	-0.06	0.17	0.12	0.06	-0.56	-0.01	1.00						
WML	1987Q1-1997Q1	41	10.17%	10.52%	0.03	-0.12	0.09	0.13	-0.26	-0.04	-0.04	-0.29	1.00					
DY	1977Q1-1997Q2	82	2.58%	0.78%	0.94	0.87	0.81	0.74	0.60	-0.51	-0.10	0.44	-0.14	1.00				
TB	1977Q1-1997Q3	83	5.94%	2.29%	0.94	0.85	0.73	0.62	0.08	0.00	-0.13	0.25	-0.12	0.28	1.00			
TERM	1977Q1-1997Q3	83	1.34%	1.42%	0.89	0.75	0.63	0.50	-0.02	-0.14	-0.04	-0.14	0.17	-0.24	-0.92	1.00		
GDP growth	1977Q1-1997Q2	82	2.59%	2.94%	-0.04	0.01	0.09	0.24	-0.07	0.05	-0.37	-0.04	0.06	0.00	0.12	-0.01	1.00	
IDP growth	1997Q1-1997Q3	83	1.22%	5.17%	-0.21	-0.05	0.08	-0.11	-0.09	-0.01	0.18	0.15	-0.06	-0.20	-0.25	0.29	0.22	1.00
Italy																		
MKT	1978Q3-1997Q1	75	8.74%	29.51%	0.07	0.24	-0.03	0.13	0.13	1.00								
HML	1987Q2-1997Q1	40	8.95%	10.37%	0.08	0.05	0.14	0.14	-0.14	0.56	1.00							
SMB	1987Q2-1997Q1	40	1.78%	10.34%	0.16	-0.08	0.10	-0.04	0.12	0.32	0.67	1.00						
WML	1987Q2-1997Q1	40	6.10%	15.31%	-0.19	-0.08	0.02	0.02	0.02	-0.41	-0.51	-0.23	1.00					
DY	1977Q1-1997Q2	82	2.79%	1.07%	0.91	0.82	0.70	0.57	0.01	-0.26	-0.32	-0.27	0.03	1.00				
TB	1977Q1-1997Q3	83	12.28%	3.94%	0.83	0.78	0.72	0.68	0.45	-0.38	-0.30	-0.41	-0.03	0.82	1.00			
TERM	1991Q2-1997Q3	26	1.65%	1.30%	0.58	0.17	-0.15	-0.17	-0.25	0.02	0.03	-0.07	-0.07	-0.36	-0.35	1.00		
GDP growth	1977Q1-1997Q3	83	2.07%	1.39%	0.34	0.18	0.05	0.02	-0.13	0.15	0.32	0.50	-0.30	-0.44	-0.53	0.30	1.00	
IDP growth	1977Q1-1997Q3	83	1.43%	6.40%	-0.11	-0.17	-0.02	-0.09	-0.15	0.16	0.24	0.44	-0.36	-0.26	-0.25	0.30	0.76	1.00

Japan																			
MKT	1977Q1-1997Q1	81	3.94%	18.26%	-0.05	0.09	0.02	-0.15	0.02	1.00									
HML	1980Q3-1997Q1	67	12.53%	9.83%	0.18	0.08	0.03	0.10	0.01	-0.19	1.00								
SMB	1980Q3-1997Q1	67	6.22%	14.71%	0.05	0.07	-0.05	0.06	0.15	0.06	-0.03	1.00							
WML	1980Q3-1997Q1	67	0.55%	10.85%	-0.10	0.13	-0.13	0.17	0.14	0.09	-0.04	-0.17	1.00						
DY	1977Q1-1997Q2	82	1.20%	0.52%	0.96	0.91	0.86	0.81	0.66	-0.03	0.09	-0.24	0.09	1.00					
TB	1977Q1-1997Q3	83	4.89%	2.51%	0.94	0.84	0.74	0.64	0.33	0.00	0.21	0.05	0.10	0.46	1.00				
TERM	1977Q1-1997Q3	83	1.08%	1.22%	0.78	0.05	0.36	0.20	-0.03	-0.04	-0.19	-0.10	0.02	0.11	-0.70	1.00			
GDP growth*	1977Q1-1997Q3	83	3.26%	1.61%	0.01	0.12	0.23	0.21	-0.07	0.02	0.20	0.23	0.05	-0.11	0.19	-0.14	1.00		
IDP growth	1977Q1-1997Q3	83	2.98%	3.26%	0.06	0.28	0.15	0.15	-0.16	0.07	-0.09	0.08	0.08	-0.17	-0.04	0.16	0.20	1.00	
Netherlands																			
MKT	1977Q1-1997Q1	81	10.80%	15.44%	0.06	-0.06	0.03	0.10	-0.02	1.00									
HML	1978Q2-1997Q1	76	0.24%	12.55%	0.31	0.09	0.04	0.00	-0.16	-0.05	1.00								
SMB	1978Q2-1997Q1	76	0.09%	13.06%	-0.18	0.07	-0.10	0.18	0.03	0.11	0.29	1.00							
WML	1978Q2-1997Q1	76	13.83%	12.65%	-0.02	0.03	0.12	0.09	-0.06	-0.16	-0.30	-0.22	1.00						
DY	1977Q1-1997Q2	82	4.94%	1.54%	0.94	0.88	0.83	0.78	0.59	-0.21	0.05	-0.07	0.00	1.00					
TB	1977Q1-1997Q3	83	6.38%	2.32%	0.90	0.77	0.69	0.58	0.14	-0.25	0.09	-0.09	-0.09	0.65	1.00				
TERM	1977Q1-1997Q3	83	1.45%	1.48%	0.77	0.58	0.50	0.39	-0.02	0.14	-0.14	0.06	0.19	-0.13	-0.72	1.00			
GDP growth	1977Q2-1997Q3	82	2.22%	1.88%	-0.12	0.00	0.05	-0.07	0.02	0.17	-0.01	-0.12	0.09	-0.28	-0.24	0.04	1.00		
IDP growth	1977Q1-1997Q3	83	1.34%	6.31%	-0.39	-0.03	-0.03	0.02	0.04	-0.06	0.04	0.10	0.07	-0.10	-0.11	0.09	-0.10	1.00	
Switzerland																			
MKT	1977Q1-1997Q1	81	7.34%	16.54%	-0.07	-0.02	0.10	-0.04	-0.07	1.00									
HML	1986Q2-1997Q1	44	6.52%	11.56%	-0.34	0.14	0.02	0.08	0.06	0.40	1.00								
SMB	1986Q2-1997Q1	44	-2.46%	10.99%	0.23	0.19	0.11	0.17	-0.11	-0.26	0.22	1.00							
WML	1986Q2-1997Q1	44	9.06%	11.68%	-0.40	0.20	-0.20	0.11	0.07	0.13	0.05	-0.17	1.00						
DY	1977Q1-1997Q2	82	2.42%	0.50%	0.90	0.80	0.73	0.66	0.43	-0.39	-0.21	-0.11	-0.11	1.00					
TB	1977Q1-1997Q3	83	4.36%	2.47%	0.90	0.77	0.63	0.49	0.03	0.06	-0.06	-0.21	0.03	0.58	1.00				
TERM	1977Q1-1997Q3	83	0.33%	1.76%	0.85	0.69	0.54	0.40	0.01	-0.14	-0.06	0.13	-0.01	-0.49	-0.95	1.00			
GDP growth	1977Q1-1996Q3	79	1.60%	1.76%	0.70	0.28	0.10	0.06	-0.25	0.04	-0.15	-0.10	0.00	-0.02	0.36	-0.25	1.00		
IDP growth	1977Q1-1997Q3	83	1.87%	5.01%	-0.25	-0.10	0.07	-0.12	-0.12	-0.19	0.11	0.02	0.10	0.05	-0.08	0.08	-0.10	1.00	
United Kingdom																			
MKT	1977Q1-1997Q1	81	9.45%	16.59%	-0.09	-0.12	-0.09	-0.11	0.16	1.00									
HML	1977Q4-1997Q1	78	8.66%	8.42%	0.50	0.23	0.15	0.10	0.08	-0.20	1.00								
SMB	1977Q4-1997Q1	78	4.68%	12.60%	0.24	0.12	0.05	0.09	-0.01	-0.15	0.24	1.00							
WML	1977Q4-1997Q1	78	15.40%	7.42%	0.11	-0.07	-0.14	0.14	-0.04	0.31	-0.39	-0.07	1.00						
DY	1977Q1-1997Q2	82	4.64%	0.88%	0.89	0.80	0.73	0.66	0.47	-0.20	-0.17	-0.32	0.07	1.00					
TB	1977Q1-1997Q3	83	9.98%	3.09%	0.91	0.82	0.73	0.63	0.24	-0.03	-0.04	-0.36	0.06	0.71	1.00				
TERM	1977Q1-1997Q3	83	0.62%	1.97%	0.81	0.68	0.52	0.36	-0.07	-0.07	0.08	0.41	-0.13	-0.21	-0.74	1.00			
GDP growth	1977Q1-1997Q2	82	2.13%	1.79%	0.12	0.35	0.27	0.09	-0.16	0.24	0.12	0.20	0.05	-0.49	-0.40	0.35	1.00		
IDP growth	1977Q1-1997Q3	83	1.50%	3.51%	0.07	0.13	-0.09	0.02	-0.23	-0.14	0.26	0.19	-0.08	-0.29	-0.25	0.28	0.27	1.00	
United States																			
MKT	1977Q1-1997Q1	81	7.30%	14.89%	0.05	-0.11	-0.09	-0.11	0.06	1.00									
HML	1977Q4-1997Q1	78	8.82%	11.14%	0.10	-0.02	-0.02	0.23	0.13	-0.39	1.00								
SMB	1977Q4-1997Q1	78	9.48%	12.79%	-0.16	0.10	0.02	0.36	0.33	0.33	-0.05	1.00							
WML	1977Q4-1997Q1	78	11.54%	8.66%	0.04	-0.05	-0.23	-0.04	0.08	0.38	-0.50	0.25	1.00						
DY	1977Q1-1997Q2	82	3.69%	1.09%	0.94	0.88	0.82	0.77	0.66	-0.16	-0.09	-0.06	0.01	1.00					
TB	1977Q1-1997Q3	83	7.73%	3.32%	0.93	0.86	0.80	0.73	0.46	-0.08	-0.13	-0.02	-0.02	0.77	1.00				
TERM	1977Q1-1997Q3	83	1.01%	2.02%	0.84	0.71	0.64	0.56	0.30	-0.03	0.27	-0.11	-0.09	-0.46	-0.80	1.00			
GDP growth	1977Q1-1997Q2	82	2.67%	1.74%	0.28	0.11	0.11	0.05	-0.24	0.18	0.03	0.17	0.20	-0.17	-0.16	0.10	1.00		
IDP growth	1977Q1-1997Q3	83	2.69%	3.17%	0.24	0.07	0.11	-0.02	-0.14	-0.25	0.10	0.11	-0.09	-0.24	-0.13	0.10	0.43	1.00	

MKT is the excess return on the local market index. HML is the return on a portfolio that is long on high book-to-market stocks and short on low book-to-market stocks, holding the size and momentum characteristics of the portfolio constant. SMB is the return on a portfolio that is long on small capitalization stocks and short on large capitalization stocks, holding the book-to-market and momentum characteristics of the portfolio constant. WML is the return on a portfolio that is long on the best performing stocks of the past year ("winners") and short on the worst performing stocks of the past year ("losers"), holding the book-to-market and size effects of the portfolio constant. TB denotes the annual Treasury Bill yield. DY is the dividend yield on a country's market capitalization weighted index. TERM stands for the premium on a ten-year government bond minus the TB rate. IDP growth is the growth in a country's industrial production. GDP growth is calculated as the continuously compounded growth rate in a country's Gross Domestic Product, which is seasonally adjusted. For Japan we use the seasonally adjusted Gross National Product. All returns are annualized and continuously compounded.

Appendix 6
Summary of Long-Short Portfolios Annual Returns
(Stocks are sorted in portfolios first according to ave12, then according to mv and finally according to b/m)

Country	Quarterly Rebalancing			Semi-Annual Rebalancing			Annual Rebalancing		
	mean	std.	t-stat.	mean	std.	t-stat.	mean	std.	t-stat.
Australia									
HML	7.72%	13.61%	2.02**	4.77%	14.96%	1.12	-1.30%	14.43%	-0.31
SMB	8.81%	15.99%	1.96*	7.98%	16.35%	1.72*	11.77%	19.05%	2.13**
WML	9.33%	17.75%	1.87*	7.39%	17.39%	1.50	5.15%	18.76%	0.95
Canada									
HML	12.88%	9.76%	5.85**	9.53%	9.61%	4.37**	3.66%	10.43%	1.65*
SMB	9.31%	10.73%	3.85**	8.75%	10.56%	3.65**	8.85%	10.03%	3.83**
WML	13.35%	13.90%	4.26**	9.54%	13.55%	3.10**	2.14%	12.93%	0.72
France									
HML	13.40%	8.67%	5.82**	10.58%	8.85%	4.46**	6.31%	9.56%	2.37**
SMB	10.10%	11.87%	3.20**	9.20%	12.27%	2.80**	8.97%	10.69%	3.02**
WML	10.52%	9.76%	4.05**	9.16%	9.25%	3.69**	5.49%	8.40%	2.35**
Germany									
HML	7.95%	6.85%	3.75**	6.25%	5.91%	3.33**	4.21%	6.88%	1.93*
SMB	3.96%	8.39%	1.52	2.11%	8.36%	0.79	1.79%	8.48%	0.66
WML	9.30%	9.77%	3.07**	8.70%	9.67%	2.83**	7.96%	8.37%	3.00**
Italy									
HML	8.90%	8.40%	3.38**	3.81%	10.58%	1.13	6.25%	9.61%	2.05**
SMB	2.95%	10.71%	0.87	3.66%	12.69%	0.91	3.52%	10.33%	1.07
WML	4.74%	13.51%	1.12	6.68%	9.10%	2.97**	-2.78%	12.95%	-0.68
Japan									
HML	9.92%	9.03%	4.52**	6.68%	9.10%	2.97**	7.84%	8.49%	3.68**
SMB	6.66%	14.13%	1.94*	6.57%	13.79%	1.93*	5.54%	13.08%	1.69*
WML	-1.65%	12.52%	-0.54	-2.66%	12.18%	-0.89	-6.02%	11.59%	-2.07*
Netherlands									
HML	-0.56%	11.26%	-0.22	-0.58%	11.31%	-0.22	-4.66%	11.41%	-1.73*
SMB	1.81%	12.34%	0.64	2.36%	12.59%	0.81	2.33%	12.76%	0.77
WML	14.81%	12.97%	5.00**	11.56%	13.01%	3.87**	10.37%	12.99%	3.38**
Switzerland									
HML	7.16%	8.26%	2.90**	6.81%	8.69%	2.59**	7.43%	7.68%	3.04**
SMB	2.16%	10.99%	0.66	2.17%	10.91%	0.66	-1.69%	9.25%	-0.58
WML	8.60%	11.04%	2.60**	5.87%	11.58%	1.67*	7.36%	9.13%	2.54**
United Kingdom									
HML	9.42%	6.01%	6.95**	6.96%	6.04%	5.07**	4.09%	5.88%	3.03**
SMB	8.29%	11.26%	3.26**	7.33%	11.29%	2.86**	6.82%	11.09%	2.68**
WML	13.15%	6.85%	8.52**	11.26%	6.47%	7.67**	7.12%	5.98%	5.18**
United States									
HML	7.85%	7.71%	4.51**	5.93%	7.97%	3.28**	3.85%	7.69%	2.18**
SMB	10.89%	10.92%	4.42**	10.45%	11.25%	4.09**	8.41%	10.39%	3.52**
WML	10.31%	10.58%	4.32**	9.47%	10.01%	4.17**	4.00%	10.45%	1.67*

* Significant at the 10% Level.

** Significant at the 5% Level.

Same comments as in Table 1 apply.

Appendix 7: The Performance of the HML, SMB, and WML Strategies at Good, Bad, and Normal States of the Economy
(Stocks are sorted in portfolios first according to ave12, then according to mv and finally according to b/m)

Past Year Return on Return-Based Factor Sorted by Future GDP Growth									
Country	HML			SMB			WML		
	Good State	Normal	Bad State	Good State	Normal	Bad State	Good State	Normal	Bad State
Australia	6.62%	7.48%	13.40%	20.39%	9.44%	0.38%	7.85%	6.28%	14.07%
Canada	10.39%	13.25%	17.05%	13.40%	12.15%	3.92%	17.97%	16.10%	9.97%
France	20.77%	15.85%	7.49%	21.18%	10.19%	1.21%	2.73%	16.51%	10.51%
Germany	8.19%	9.16%	4.69%	10.87%	4.52%	0.11%	18.13%	7.30%	11.16%
Italy	14.72%	8.51%	5.89%	5.45%	5.09%	-4.80%	-1.09%	5.58%	5.37%
Japan	11.00%	11.88%	8.26%	12.54%	9.74%	2.99%	-0.66%	0.43%	-7.26%
Netherlands	4.63%	-2.78%	-1.43%	8.76%	-0.45%	1.60%	8.23%	15.43%	20.22%
Switzerland	10.29%	7.80%	3.23%	11.11%	-3.35%	3.61%	17.13%	6.26%	8.79%
United Kingdom	16.96%	8.48%	6.23%	19.28%	11.21%	-5.53%	12.88%	12.32%	17.47%
United States	8.32%	10.96%	3.90%	15.81%	9.37%	9.79%	12.19%	10.34%	9.91%

Same comments as in Table 2 apply.

Appendix 8: Univariate Regressions of GDP growth rates
Conditional on Past One-Quarter, Two-Quarters, and Four-Quarters of Factor Returns
 $GDPgrowth(t,t+4) = a + b * FactorRet(t-n,t) + e(t,t+n)$ where $n=1,2$, or 4 .
(Stocks are sorted in portfolios first according to ave12, then according to mv and finally according to b/m)

Country	Time Period	Slope Coefficient				t-statistics				Coefficient of Determinations			
		MKT	HML	SMB	WML	MKT	HML	SMB	WML	MKT	HML	SMB	WML
Past One Quarter													
GDPgrowth(t,t+4) = a + b *FactorRet (t-1,t) + e(t,t+4)													
Australia	1984Q4-1996Q2	0.047	-0.025	0.057	-0.050	1.28	-0.62	2.06**	-1.94*	3.9%	-1.1%	5.4%	0.9%
Canada	1977Q4-1996Q2	0.133	-0.024	0.052	-0.029	3.03**	-0.41	1.79	-0.60	15.7%	-1.2%	0.2%	-0.6%
France	1983Q2-1996Q2	0.016	0.045	0.099	-0.021	0.73	1.21	3.23**	-0.67	-0.7%	-0.2%	12.4%	-1.2%
Germany	1987Q1-1996Q3	-0.011	-0.228	0.259	0.049	-0.17	-1.42	2.32**	0.39	-2.6%	3.9%	6.8%	-2.2%
Italy	1987Q2-1996Q3	0.031	0.096	0.117	-0.026	1.19	1.77*	2.76**	-0.87	2.0%	4.5%	12.4%	-1.2%
Japan	1980Q3-1996Q3	0.018	0.023	0.046	0.017	0.72	0.78	1.57	0.52	-0.6%	-1.2%	2.0%	-1.3%
Netherlands	1978Q2-1996Q3	0.059	0.012	-0.019	0.001	2.17**	0.36	-0.55	0.02	5.3%	-1.2%	-0.9%	-1.4%
Switzerland	1986Q2-1995Q4	0.010	0.042	-0.018	-0.012	0.30	0.76	-0.42	-0.56	-2.3%	-1.2%	-2.2%	-2.3%
United Kingdom	1977Q4-1996Q3	0.070	0.141	0.137	-0.073	1.95*	1.82*	2.88**	-1.47	4.5%	3.3%	12.8%	0.2%
United States	1977Q4-1996Q3	0.084	-0.007	0.079	0.061	2.01**	-0.11	1.97**	0.92	7.2%	-1.3%	4.5%	0.6%
Past Two-Quarters													
GDPgrowth(t,t+4) = a + b *FactorRet (t-2,t) + e(t,t+4)													
Australia	1985Q1-1996Q2	0.048	-0.021	0.055	-0.066	1.47	-0.67	2.09**	-1.61	9.8%	-0.6%	11.4%	5.5%
Canada	1978Q1-1996Q2	0.098	-0.051	0.047	-0.005	3.28**	-0.87	2.54**	-0.15	23.0%	0.6%	1.8%	-1.3%
France	1983Q3-1996Q2	0.014	0.065	0.084	-0.030	0.82	2.37**	4.22**	-1.25	0.3%	6.8%	22.7%	1.5%
Germany	1987Q2-1996Q3	0.002	-0.111	0.244	0.086	0.03	-0.84	3.07**	0.80	-2.8%	0.2%	15.3%	1.1%
Italy	1987Q3-1996Q3	0.031	0.114	0.088	-0.020	1.44	2.77**	3.54**	-0.92	7.8%	17.3%	18.7%	-1.0%
Japan	1980Q4-1996Q3	0.024	0.007	0.038	0.023	1.16	0.29	1.75*	0.86	2.1%	-1.6%	3.7%	-0.4%
Netherlands	1978Q3-1996Q3	0.046	0.013	0.003	-0.010	2.41**	0.51	0.10	-0.33	8.1%	-1.0%	-1.4%	-1.1%
Switzerland	1986Q3-1995Q4	0.014	0.069	-0.022	-0.006	0.47	1.12	-0.57	-0.22	-1.2%	3.4%	-0.8%	-2.7%
United Kingdom	1978Q1-1996Q3	0.064	0.106	0.114	-0.054	2.29**	2.03**	3.67**	-1.56	8.6%	6.5%	23.7%	0.9%
United States	1978Q1-1996Q3	0.069	0.001	0.067	0.049	2.23**	0.03	1.90*	0.91	12.7%	-1.4%	6.9%	1.2%
Past Four-Quarters													
GDPgrowth(t,t+4) = a + b *FactorRet (t-4,t) + e(t,t+4)													
Australia	1985Q3-1996Q2	0.028	-0.020	0.045	-0.049	1.31	-1.10	2.62**	-1.27	7.5%	0.8%	23.4%	6.3%
Canada	1978Q3-1996Q2	0.051	-0.067	0.041	0.015	2.93**	-1.45	3.11**	0.72	16.3%	5.9%	5.0%	0.0%
France	1984Q1-1996Q2	0.009	0.053	0.059	-0.021	0.77	2.57**	3.84**	-1.12	0.1%	15.5%	24.3%	2.1%
Germany	1987Q4-1996Q3	0.056	0.017	0.150	0.112	1.40	0.24	2.39**	1.30	6.4%	-2.8%	7.9%	12.4%
Italy	1988Q1-1996Q3	0.031	0.087	0.052	-0.022	2.75**	2.71**	2.31**	-1.56	25.4%	23.2%	15.0%	1.3%
Japan	1981Q2-1996Q3	0.031	0.018	0.020	0.049	2.04**	0.87	1.39	2.43**	12.3%	-0.7%	1.7%	8.3%
Netherlands	1979Q1-1996Q3	0.029	0.011	0.010	-0.021	1.90*	0.44	0.34	-1.14	7.6%	-1.0%	-1.0%	1.3%
Switzerland	1987Q1-1995Q4	0.016	0.072	-0.033	0.023	0.86	1.46	-0.91	1.10	1.4%	11.9%	7.1%	0.2%
United Kingdom	1978Q3-1996Q3	0.069	0.072	0.087	-0.054	3.29**	2.06**	4.51**	-1.52	19.3%	9.5%	32.8%	2.9%
United States	1978Q3-1996Q3	0.047	0.024	0.047	0.048	2.33**	0.79	1.42	1.42	11.8%	-0.1%	6.8%	5.7%

* Significant at the 10% Level.

** Significant at the 5% Level.

Same comments as in Table 3 apply.

Appendix 9: Predicting Annual GDP Growth Rates Conditional on Information about the Return on the Market and HML, SMB, and WML Trading Strategies : Bivariate Regressions

$$\text{GDPgrowth}(t,t+4) = a + b \cdot \text{MKT}(t-4,t) + c \cdot \text{Factor} \cdot \text{Ret}(t-4,t) + e(t,t+4)$$

(Stocks are sorted in portfolios first according to ave12, then according to mv and finally according to b/m)

Country	Time Period	MKT		HML		adj. R2
		slope	t-stat.	slope	t-stat.	
Australia	1985Q3-1996Q2	0.028	1.14	0.000	0.00	5.23%
Canada	1978Q3-1996Q2	0.046	2.72**	-0.049	-1.10	18.96%
France	1984Q1-1996Q2	-0.009	-0.60	0.063	2.22**	15.04%
Germany	1987Q4-1996Q3	0.062	1.43	-0.036	-0.41	4.18%
Italy	1988Q1-1996Q3	0.022	1.90*	0.056	1.68*	31.86%
Japan	1981Q2-1996Q3	0.031	2.18**	0.022	1.03	12.23%
Netherlands	1979Q1-1996Q3	0.035	2.58**	0.030	1.32	9.40%
Switzerland	1987Q1-1995Q4	-0.004	-0.23	0.078	1.45	9.43%
United Kingdom	1978Q3-1996Q3	0.071	3.95**	0.076	2.64**	30.48%
United States	1978Q3-1996Q3	0.062	2.92**	0.061	2.44**	18.28%

Country		MKT		SMB		adj. R2
		slope	t-stat.	slope	t-stat.	
Australia	1985Q3-1996Q2	0.019	1.04	0.041	2.75**	25.65%
Canada	1978Q3-1996Q2	0.051	3.19**	0.041	2.75**	21.57%
France	1984Q1-1996Q2	-0.001	-0.10	0.060	3.72**	22.76%
Germany	1987Q4-1996Q3	0.089	2.00**	0.228	2.57**	26.20%
Italy	1988Q1-1996Q3	0.027	2.70**	0.040	2.67**	33.85%
Japan	1981Q2-1996Q3	0.030	1.96*	0.016	1.24	13.04%
Netherlands	1979Q1-1996Q3	0.033	2.51**	0.021	0.78	8.39%
Switzerland	1987Q1-1995Q4	0.015	0.84	-0.032	-0.96	8.07%
United Kingdom	1978Q3-1996Q3	0.045	2.61**	0.073	4.46**	39.91%
United States	1978Q3-1996Q3	0.039	2.19**	0.029	0.97	13.14%

Country		MKT		WML		adj. R2
		slope	t-stat.	slope	t-stat.	
Australia	1985Q3-1996Q2	0.031	1.75*	-0.053	-1.52	15.92%
Canada	1978Q3-1996Q2	0.050	2.89**	0.011	0.69	15.84%
France	1984Q1-1996Q2	0.013	0.14	-0.031	-1.73*	5.99%
Germany	1987Q4-1996Q3	0.032	0.88	0.092	1.08	12.46%
Italy	1988Q1-1996Q3	0.030	2.60**	-0.006	-0.55	23.36%
Japan	1981Q2-1996Q3	0.026	1.51	0.035	1.40	15.60%
Netherlands	1979Q1-1996Q3	0.033	2.24**	-0.028	-1.85*	11.05%
Switzerland	1987Q1-1995Q4	0.014	0.71	0.017	0.91	0.12%
United Kingdom	1978Q3-1996Q3	0.069	3.70**	-0.052	-1.50	22.18%
United States	1978Q3-1996Q3	0.043	2.12**	0.016	0.41	10.95%

* Significant at the 10% Level.

** Significant at the 5% Level.

Same comments as in Table 4 apply.

Appendix 10: Predicting Annual GDP Growth Rates Conditional on Information about the Return on the Market and HML, SMB, and WML Trading Strategies :
Multivariate Regressions

$$\text{GDPgrowth}(t,t+4) = a + b \cdot \text{MKT}(t-4,t) + c \cdot \text{HML}(t-4,t) + d \cdot \text{SMB}(t-4,t) + e(t,t+4)$$

(Stocks are sorted in portfolios first according to ave12, then according to mv and finally according to b/m)

Country	Time Period	MKT		HML		SMB		adj. R2
		slope	t-stat.	slope	t-stat.	slope	t-stat.	
Australia	1985Q3-1996Q2	0.014	0.66	-0.009	-0.47	0.042	2.66**	24.28%
Canada	1978Q3-1996Q2	0.046	2.95**	-0.055	-1.22	0.044	2.90**	25.18%
France	1984Q1-1996Q2	-0.013	-1.03	0.048	1.93*	0.051	3.63**	30.55%
Germany	1987Q4-1996Q3	0.088	1.89*	0.014	0.21	0.231	2.68**	23.98%
Italy	1988Q1-1996Q3	0.022	1.99**	0.038	1.10	0.031	1.87*	35.14%
Japan	1981Q2-1996Q3	0.030	2.11**	0.027	1.24	0.019	1.48	13.81%
Netherlands	1979Q1-1996Q3	0.036	2.90**	0.024	0.89	0.014	0.45	8.82%
Switzerland	1987Q1-1995Q4	-0.015	-0.74	0.112	1.72*	-0.047	-1.63	26.19%
United Kingdom	1978Q3-1996Q3	0.050	3.05**	0.051	1.90*	0.062	3.49**	44.27%
United States	1978Q3-1996Q3	0.053	2.88**	0.065	2.49**	0.033	1.20	20.66%

* Significant at the 10% Level.

** Significant at the 5% Level.

Same comments as in Table 5 apply.

Appendix 11: The Ability of Stock Market Factors to Predict Annual GDP Growth in the Presence of Business Cycle Variables

$$\text{GDPgrowth}(t,t+4) = a + b*\text{MKT}(t-4,t) + c*\text{FactorRet}(t-4,t) + d*\text{TB}(t) + f*\text{DY}(t) + g*\text{TERM}(t) + h*\text{IDPgrowth}(t-4,t) + e(t,t+4)$$

(Stocks are sorted in portfolios first according to ave12, then according to mv and finally according to b/m)

Country	Time Period	MKT		HML		TB		DY		TERM		IDP growth		adj. R2
		slope	t-stat.	slope	t-stat.	slope	t-stat.	slope	t-stat.	slope	t-stat.	slope	t-stat.	
Australia	1985Q3-1995Q2	0.051	-5.22**	-0.010	-0.76	-0.276	-2.75**	-3.108	-8.44	-0.634	-4.05**	-0.093	-1.61	73.0%
Canada	1978Q3-1996Q2	0.035	2.67**	-0.075	-2.15**	-0.033	-0.20	0.401	0.72	0.744	3.19**	0.105	1.57	53.1%
France	1984Q1-1996Q2	0.016	-1.79*	0.075	3.23**	-0.224	-1.15	-0.607	-2.00**	0.000	0.00	0.076	1.21	31.8%
Germany	1987Q4-1996Q3	0.136	2.42**	0.072	1.08	2.525	1.96**	5.807	1.81*	3.226	2.08**	0.157	1.09	36.1%
Italy	1988Q1-1996Q3	0.031	3.58**	0.011	0.46	-0.261	-1.43	-0.079	-0.17	0.196	1.47	0.002	0.04	70.2%
Japan	1981Q2-1996Q3	0.021	1.85*	0.030	1.94*	0.215	0.79	-0.942	-0.93	0.279	0.77	0.246	4.31**	49.7%
Netherlands	1979Q1-1996Q3	0.016	1.52	0.029	1.43	-0.876	-3.79**	0.305	1.10	-0.658	-2.38**	0.036	0.92	44.5%
Switzerland	1987Q1-1995Q4	-0.017	-1.15	0.064	2.07**	1.161	3.77**	-2.247	-1.93*	1.124	3.26**	0.017	0.25	43.4%
United Kingdom	1978Q3-1996Q3	0.032	1.76*	0.086	3.70**	0.011	0.06	-0.936	-1.39	0.701	3.44**	-0.060	-0.81	62.5%
United States	1978Q3-1996Q3	0.060	2.59**	-0.015	-0.56	0.006	0.03	0.696	1.50	0.703	3.56**	0.049	0.80	40.8%

Country	Time Period	MKT		SMB		TB		DY		TERM		IDP growth		adj. R2
		slope	t-stat.	slope	t-stat.	slope	t-stat.	slope	t-stat.	slope	t-stat.	slope	t-stat.	
Australia	1985Q3-1995Q2	-0.048	-6.18**	0.006	0.45	-0.185	-1.10	-3.138	-8.71**	-0.504	-1.88*	-0.095	-2.05**	72.7%
Canada	1978Q3-1996Q2	0.053	4.17**	-0.004	-0.29	-0.111	-0.51	0.776	1.26	0.769	3.08**	0.016	0.25	48.4%
France	1984Q1-1996Q2	-0.008	-0.82	0.063	4.22**	-0.058	-0.33	-0.949	-2.17**	0.133	0.48	-0.117	-2.04**	35.0%
Germany	1987Q4-1996Q3	0.189	2.56**	0.211	1.99**	2.480	2.27**	5.694	1.62	3.779	2.60**	-0.147	-0.76	45.1%
Italy	1988Q1-1996Q3	0.030	4.23**	0.022	2.07**	-0.221	-1.18	-0.317	-0.54	0.197	1.50	-0.041	-0.69	71.6%
Japan	1981Q2-1996Q3	0.225	2.07**	-0.022	-1.74*	0.242	0.85	-1.219	-1.10	0.130	0.34	0.259	4.50**	49.9%
Netherlands	1979Q1-1996Q3	0.013	1.52	0.026	1.75*	-0.856	-3.69**	0.236	0.84	-0.626	-2.39**	0.021	0.50	44.8%
Switzerland	1987Q1-1995Q4	-0.006	-0.41	-0.007	-0.26	1.181	3.67**	-3.058	-2.22**	0.991	2.56**	0.091	1.09	38.6%
United Kingdom	1978Q3-1996Q3	0.015	0.71	0.036	2.13**	0.159	0.84	-1.424	-1.67*	0.593	2.29**	-0.038	-0.39	55.0%
United States	1978Q3-1996Q3	0.049	2.83**	0.042	1.91*	0.062	0.33	0.532	1.36	0.745	4.10**	0.007	0.11	46.0%

Country	Time Period	MKT		WML		TB		DY		TERM		IDP growth		adj. R2
		slope	t-stat.	slope	t-stat.	slope	t-stat.	slope	t-stat.	slope	t-stat.	slope	t-stat.	
Australia	1985Q3-1995Q2	-0.051	-4.42**	0.009	0.42	-0.225	-1.90*	-3.270	-6.66**	-0.552	-2.55**	-0.098	-1.98**	72.6%
Canada	1978Q3-1996Q2	0.051	4.34**	0.010	0.76	-0.080	-0.41	0.792	1.28	0.778	3.12**	0.034	0.49	48.8%
France	1984Q1-1996Q2	0.005	0.49	-0.022	-1.59	0.141	0.79	-1.111	-2.65**	0.346	1.35	-0.010	-0.16	19.2%
Germany	1987Q4-1996Q3	0.110	2.13**	0.063	1.34	1.615	1.59	5.194	1.72*	1.967	1.48	0.266	1.67*	37.7%
Italy	1988Q1-1996Q3	0.044	3.09**	0.021	1.17	-0.012	-0.07	-0.411	-0.95	0.428	2.35**	0.010	0.21	70.5%
Japan	1981Q2-1996Q3	0.019	1.73*	0.015	0.72	0.178	0.62	-0.832	-0.78	0.155	0.43	0.234	4.03**	47.9%
Netherlands	1979Q1-1996Q3	0.012	1.27	-0.020	-1.51	-0.828	-3.48**	0.274	1.03	-0.586	-1.98**	0.036	0.88	43.8%
Switzerland	1987Q1-1995Q4	-0.004	-0.30	-0.007	-0.64	1.247	3.91**	-3.051	-2.47**	1.055	3.09**	0.094	1.29	38.6%
United Kingdom	1978Q3-1996Q3	0.014	0.61	0.023	0.76	0.036	1.66*	-2.038	-2.43**	0.937	3.38**	0.002	0.02	52.3%
United States	1978Q3-1996Q3	0.044	2.13**	0.057	2.40**	0.124	0.79	0.399	0.98	0.792	4.21**	0.088	1.53	44.5%

* Significant at the 5% level.

** Significant at the 10% level.

Same comments as in Table 6A apply.

Appendix 12: The Ability of Stock Market Factors to Predict Annual GDP Growth in the Presence of Business Cycle Variables

$$\text{GDPgrowth}(t,t+4) = a + b*\text{MKT}(t-4,t) + c*\text{HML}(t-4,t) + d*\text{SMB}(t-4,t) + f*\text{TB}(t) + g*\text{DY}(t) + h*\text{TERM}(t) + i*\text{IDPgrowth}(t-4,t) + e(t,t+4)$$

(Stocks are sorted in portfolios first according to ave12, then according to inv and finally according to b/m)

Country	Time Period	MKT		HML		SMB		TB		DY		TERM		IDP growth		adj. R2
		slope	t-stat.	slope	t-stat.	slope	t-stat.	slope	t-stat.	slope	t-stat.	slope	t-stat.	slope	t-stat.	
Australia	1985Q3-1995Q2	-0.051	-5.43**	-0.009	-0.75	0.005	0.37	-0.229	-1.32	-3.084	-8.84**	-0.572	-2.30**	-0.085	-1.65*	72.1%
Canada	1978Q3-1996Q2	0.035	2.62**	-0.075	-2.15**	0.000	0.02	-0.032	-0.18	0.400	0.71	0.744	3.22**	0.106	1.56	52.3%
France	1984Q1-1996Q2	-0.021	-2.38**	0.064	3.70**	0.056	4.39**	-0.322	-2.31**	-0.486	-1.85*	-0.122	-0.59	-0.038	-0.73	45.6%
Germany	1987Q4-1996Q3	0.181	2.89**	0.139	1.96*	0.260	2.71**	3.273	2.99**	5.196	1.66*	4.670	3.40**	-0.266	-1.58	49.3%
Italy	1988Q1-1996Q3	0.034	4.08**	-0.044	-1.11	0.045	2.23**	-0.177	-1.24	-0.788	-1.62*	0.171	1.42	-0.074	-1.35	70.6%
Japan	1981Q2-1996Q3	0.022	2.06**	0.023	1.31	-0.017	-1.34	0.244	0.89	-1.234	-1.16	0.229	0.58	0.260	4.69**	50.5%
Netherlands	1979Q1-1996Q3	0.017	1.71*	0.021	1.06	0.020	1.43	-0.879	-3.95**	0.286	1.02	-0.648	-2.64**	0.025	0.68	45.3%
Switzerland	1987Q1-1995Q4	-0.027	-1.36	0.094	1.87*	-0.026	-0.88	1.016	3.62**	-2.214	-1.98**	1.068	2.89**	0.001	0.02	45.1%
United Kingdom	1978Q3-1996Q3	0.029	1.51	0.081	3.57**	0.027	1.67*	-0.069	-0.41	-0.705	-1.03	0.526	2.54**	-0.084	-1.07	63.7%
United States	1978Q3-1996Q3	0.047	2.57**	-0.010	-0.36	0.042	1.88*	0.069	0.38	0.537	1.36	0.777	4.58**	0.013	0.20	45.3%

* Significant at the 10% Level.

** Significant at the 5% Level.

Same comments as in Table 6B apply.

Endnotes

ⁱ Hawawini and Keim (1995) provide a survey of the size effect in international markets. They report a size effect in Belgium, Japan, Spain, and the United Kingdom, whereas no effect was found in Canada and France. Suggested evidence of an international size effect is also found in Heston, Rouwenhorst, and Wessels (1995).

ⁱⁱ The greater profitability of momentum strategies with shorter re-balancing intervals is consistent with Rouwenhorst's (1998) findings.

ⁱⁱⁱ In the U.S. and the UK the number of dead stocks was over 8,000. We therefore screened the defunct stocks and included only those with at least 60 price observations. The price series of defunct stocks repeat the last price. To mitigate this problem, we deleted any single price that was repeated at least six times.

^{iv} Portfolio construction follows those of Fama and French (1993), Carhart, Krail, Stevens, and Welch (1995), and Carhart (1997).

^v Excluding the most recent month return eliminates problems associated with micro-structure issues such as the bid-ask bounce; see Asness (1995).

^{vi} In addition, we reversed the sorting order of the zero-investment portfolios from (1) B/M, (2) MV, and (3) ave12 to (1) ave12, (2) MV, (3) B/M. The results obtained from using the "reversed" portfolios are contained in Appendices 6-12. The main results of the paper remain unchanged, although the results for countries with few securities in each portfolio, such as Italy, the Netherlands, and Switzerland, appear to be less robust to changes in the sorting order.

^{vii} There exist some discrepancies between our results and those of Rouwenhorst (1998). Note that Rouwenhorst's data cover the period from 1980-1995, while ours run from 1977 to 1997:5, depending on the country. Furthermore, Rouwenhorst's portfolios are equally weighted whereas ours are value weighted.

^{viii} All GDP and IPD series are seasonally adjusted. For Japan we use the GNP since the GDP was not available.

^{ix} Similar results are obtained when the order in portfolio formation is reversed see Appendix 7.

^x Our results remain qualitatively the same when the parameter q in the Newey-West estimator takes the alternative values of 4, 5, or 6.

^{xi} Examination of the industry breakdown of the quarterly and semi-annually rebalanced extreme portfolios produced similar results.